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The incremental information in the yield curve about future interest rate risk ☆

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ABSTRACT

Using high-frequency intraday futures prices to measure yield volatility at selected maturities, we find that daily yield curves carry incremental information about future interest rate risk at the long end, relative to that contained in the time series of historical volatilities. Some of the information in the yield curves is not captured by standard affine models. Our results point to the existence of an unspanned stochastic volatility factor. Both time series and yield curve based forecasts provide utility to a risk averse investor, relative to a random walk. Information from the two sources can be combined to enhance yield volatility forecasting performance.

1. Introduction

Affine term structure models have been the workhorse in the interest rate literature for decades and can successfully explain bond prices (see Duffee (2002), Christensen et al. (2011), and many others). However, it is questionable whether these models are able to explain yield volatilities, and hence reinvestment rate risk, which plays a crucial role in portfolio allocation, market timing, derivative pricing, and risk management. The literature has primarily focused on whether volatilities are spanned by interest rates, as implied by standard affine models. Collin-Dufresne et al. (2009) and Andersen and Benzoni (2010) conclude that they are not, while Jacobs and Karoui (2009) and Bikbov and Chernov (2009) reach mixed conclusions.

In this paper, we consider out-of-sample forecasting of yield volatility. Since future reinvestment rate risk matters for pricing and trading decisions, the yield curve should be sensitive to, and hence informative about, future yield volatility. Nevertheless, the forecasting perspective has so far received little attention in the yield volatility literature. The only exceptions are Collin-Dufresne et al. (2009) and Joslin and Konchitchki (2018), who consider fixed-window estimation, and do not compare the model based forecasts against time series benchmarks beyond a random walk (RW). In contrast, we update the parameters recursively according to investor's information set, and compare the model based volatility forecasts to leading time series benchmarks, in particular, the heterogeneous autoregressive (HAR) model of Corsi (2009) and the realized GARCH model of Hansen et al. (2012).

Our focus is on whether the yield curve contains incremental information about future yield volatility, beyond that available from the time series of historical volatilities. In addition, we examine whether the time series based forecasters contain incremental information about future volatility, beyond the information that can be read off the yield curve. Further, we supplement the existing static analysis of the spanning issue with a dynamic out-of-sample assessment.

As a separate contribution, we use high-frequency intraday yield curves in the analysis, thus adding precision to volatility measurements. The past two decades have seen important advances in high-frequency

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data analysis, primarily focusing on equities, e.g., Barndorff-Nielsen and Shephard (2002), Andersen et al. (2003), and Zhang et al. (2005). However, intraday data have seen little application in the interest rate literature. Notable exceptions are Faust et al. (2007), who use high-frequency data to study macroeconomic announcement effects, not volatility, and Andersen and Benzoni (2010) and Cieslak and Povala (2016), who use realized volatilities of interest rates based on a 10-minute sampling frequency, chosen to mitigate the effects of market microstructure noise. We overcome the latter issue in a more direct manner by relying on the pre-averaged realized variance estimator of Jacod et al. (2009), a noise-robust estimator of volatility with excellent empirical properties for various asset classes, as documented by Christensen et al. (2014). This choice allows us to use data at higher frequency. From US Treasury and Eurodollar futures prices, we construct yields with maturities 6 months, and 1, 5, and 7 years, at the 1-minute frequency. From these, we calculate a daily time series of realized volatility measures at each of the four maturities. We consider daily yield curve based forecasters of volatility over the subsequent month, constructed from either (i) affine term structure models, (ii) principal component analysis (PCA), or (iii) interest rate (yield and forward) spreads, estimated recursively in a standard daily yield panel. The alternative forecasters based on the time series of historical volatilities (RW, HAR, realized GARCH) serve as benchmarks.

We find that the HAR model provides a strong benchmark for yield volatility forecasting. It provides relatively accurate forecasts across all maturities considered. At the intermediate 5 year maturity, a simple forward spread, motivated by risk premium considerations, or a common factor approach provide equally accurate forecasts, and all three improve significantly over the naive random walk volatility forecast. However, we also find that the yield curve contains incremental information about future yield volatility relative to the time series of historical volatilities, including HAR, in terms of forecasting accuracy at the long end of the curve. Next, we carry out a specification test as in Cieslak and Povala (2016) and find that none of the affine or time series models considered subsumes all the relevant information about future volatility contained in the yield curve across all maturities. This indicates that the yield curve contains some important information about future reinvestment rate risk that neither the term structure models nor the volatility history capture. Extending this idea, we develop a test of whether past volatility forecast errors are informative about future forecast errors, as would be the case under unspanned stochastic volatility (USV) in the sense of Collin-Dufresne and Goldstein (2002). We perform the test by extracting a common factor from the volatility forecast errors across maturities and find that it provides significant information about future forecast errors, thus indicating the presence of USV features in the data.

We investigate the economic value of the forecasters to an investor in a portfolio allocation exercise, using a utility-based framework as in Bollerslev et al. (2018). From the results, the information about future interest rate risk contained in either the yield curve or historical volatility provides economic value to a risk averse investor. A robustness analysis shows that higher-order PCA factors beyond the first three usually explaining yields in fact carry information about volatility. The portion of future volatility information in the yield curve that is not captured by standard affine term structure models is contained in these higher-order factors rather than in nonlinearities. A comparison with forecasts of future interest rate risk extracted from a wide cross section of coupon bonds confirms that the yield curve based forecasts are well represented by the daily panel estimates. Finally, we demonstrate that information from the yield curve and the volatility history can be fruitfully combined to enhance yield volatility forecasting performance.

Our work on out-of-sample interest rate risk complements several strands of literature. Within the class of affine term structure models, conditional variances are affine in state variables. In standard versions, volatilities can be extracted from the cross section of interest rates. Under USV, some factors affect volatility, but not interest rates directly. Jacobs and Karoui (2009) examine the ability of three-factor affine models to fit conditional volatilities in-sample, using estimates from an EGARCH model to proxy for true volatility. They reach mixed conclusions, depending on the sample considered. Correlations between model-implied volatility and the proxy range between 60% and 75% in US Treasury data, but are much lower in swap data, even negative for some maturities. Collin-Dufresne et al. (2009) consider affine models with and without a USV factor and conclude that volatility cannot be extracted from the cross section of interest rates based on swap data. Andersen and Benzoni (2010) find low R^2s for regressing future variances on PCA factors from the yield curve, and argue that they conduct a test of whether variances are spanned by bonds. The validity of this regression test of USV is questioned by Bikbov and Chernov (2009), who show that low R^2 is expected if yields are observed with error, a standard assumption when estimating term structure models. Bikbov and Chernov (2009), Joslin (2017), and Joslin and Konchitchki (2018) consider both swaps and swaptions and reject the restrictions needed for affine models to generate USV. We contribute to this literature by considering the spanning question in an out-of-sample framework, and providing evidence pointing to the existence of a USV factor.

The USV puzzle has recently led to the construction of more complex models, with a focus on the ability to fit volatilities and price swaptions. Feldhütter et al. (2016) propose a nonlinear model that is able to generate features consistent with USV. Cieslak and Povala (2016) construct a model with volatility factors driven by a Wishart process. Filipović et al. (2017) present a nonlinear model featuring USV that can price both bonds and swaptions successfully. None of these papers considers out-of-sample forecasting. It is beyond the scope of the present paper to include these more complex models, as they are very time-consuming to implement and reestimate recursively.

The paper is organized as follows. Section 2 describes the forecasting methods, including the term structure models, PCA and interest rate spread based forecasts, and time series models. Section 3 describes the high-frequency futures data and the construction of realized yield volatility measures. Section 4 presents the estimation method for the term structure models based on the Kalman filter. Section 5 contains the empirical analysis, and Section 6 the robustness analysis. Section 7 concludes. The Appendix contains additional material on data, models, estimation, and results.

2. Interest rate risk forecasters

Throughout, y_t^τ denotes the maturity τ yield at time t. In our empirical out-of-sample analysis, we consider daily forecasting of month-ahead interest rate risk by maturity. To assess the forecasts, we use the high-frequency data to calculate daily realized yield volatility, labeled V_t^τ (see Section 3.4 – we consider the pre-averaged realized variance and henceforth use the terms realized volatility and realized variance interchangeably for this). In line with the volatility literature (e.g., Patton and Sheppard (2015) and Bollerslev et al. (2016)), the target for the forecast built at t, based on investor's information set, is not the realized volatility at $t+h$, but instead the aggregated realized measure $V_{t+1\to t+h}^\tau = \sum_{i=1}^h V_{t+i}^\tau$, a proxy for integrated volatility from t through $t+h$, or $IV_{t\to t+h}^\tau$, say.[1] All forecasts considered are recursive, i.e., only data through t are used to construct the forecast of $V_{t+1\to t+h}^\tau$, for each maturity τ and method considered, with h set to 22 trading days for month-ahead forecasting.[2] Methods utilizing the cross-sectional information in the yield curve are presented first, followed by methods relying exclusively on the information in the time series of historical volatilities.

[1] If the yield follows the Itô process $dy_t^\tau = \mu_t^\tau dt + \sigma_t^\tau dW_t$, then $IV_{t\to t+h}^\tau = \int_t^{t+h} \left(\sigma_s^\tau\right)^2 ds$.

[2] Results for $h = 44$, forecasting two months ahead, are in the Appendix.

Table 1
Affine models.

Model	Reference	$A_m(d)$-notation	Market price of risk
Vasicek	Vasicek (1977)	$A_0(1)$	λ
CIR	Cox et al. (1985)	$A_1(1)$	$\lambda/\sigma\sqrt{r_t}$
AFNS_0	Christensen et al. (2011)	$A_0(3)$	$(\Sigma\sqrt{S(X_t)})^{-1}(\kappa(\theta - X_t) - \tilde{\kappa}(\tilde{\theta} - X_t))$
AFNS_3	Christensen et al. (2010)	$A_3(3)$	$(\Sigma\sqrt{S(X_t)})^{-1}(\kappa(\theta - X_t) - \tilde{\kappa}(\tilde{\theta} - X_t))$

This table presents the reference, notation of Dai and Singleton (2000), and market price of risk specification for the affine term structure models considered.

2.1. Yield curve based volatility forecasters

We consider an $N \times T$ fixed-maturity panel, i.e., the yields at t are $y_t = (y_t^{\tau_1}, \dots, y_t^{\tau_N})'$, $t = 1, \dots, T$, and yields of all maturities can be used to generate volatility forecasts at a given maturity τ. For each candidate forecasting method, recursive volatility forecasts are constructed using the regression

$$V^{\tau}_{t+1\to t+h} = \rho_0^{\tau,h} + \rho_1^{\tau,h\prime} Z_t + u^{\tau,h}_{t+h}, \tag{1}$$

for fixed τ, h, with $\rho_1^{\tau,h}$ a $q \times 1$ vector of predictive coefficients, $\rho_0^{\tau,h}$ the scalar intercept, and Z_t the relevant information variable extracted from the yield panel and conditioning the forecast as of t for the given method, e.g., a q-vector of fitted PCA factors, or a conditional volatility forecast from an affine model. When constructing the forecast at t' of $V^{\tau}_{t'+1\to t'+h}$, only data through t' are used to extract $Z_1, \dots, Z_{t'}$ from the yield panel, then estimate Eq. (1) over $t = 1, \dots, t' - h$, and the forecast is $\hat{\rho}_0^{\tau,h} + \hat{\rho}_1^{\tau,h\prime} Z_{t'}$. We consider methods for extracting Z_t based on affine term structure models, PCA, common factors, and risk premiums.

The recursive regressions in Eq. (1) are used in the construction of volatility forecasts, for two reasons. First, investor is not basing the forecast of $V^{\tau}_{t'+1\to t'+h}$ exclusively on $Z_{t'}$ from the daily yields, having observed the history of realized volatilities through t'. Second, in the implementation, volatility forecasts based on Z_t include estimation error, whereas realized yield variance used as forecasting target in the assessment includes measurement error, thus leading to potential biases. Therefore, the Mincer and Zarnowitz (1969) type predictive regressions in Eq. (1) are used to leverage any indication in investor's information set that the centering or scale of Z_t from the given method do not line up well with subsequent realized volatilities. The main question is whether Z_t serves as a useful information variable, i.e., whether variation in this provides incremental predictive power. If, for example, a combination of the most recent realized volatility observations $V^{\tau}_{t'}, V^{\tau}_{t'-1}, \dots$ provides the best forecast at t', as in the HAR model, then this feature will not be fully captured by the intercept and slope in Eq. (1), estimated over the full window $t = 1, 2, \dots, t' - h$. It should be possible to improve on the resulting forecast using pure time series methods and historical volatilities, hence revealing that Z_t from the panel does not carry incremental information. Thus, the regression approach secures a level playing field for comparison of yield curve and time series based yield volatility forecasts.

In our empirical work, we use $N = 8$ maturities to extract Z_t from daily data, and volatility forecasts based on investor's information set are constructed using the recursive regressions in Eq. (1) for each of the four maturities τ for which we have high-frequency data.

2.1.1. Affine term structure models

In the affine class, the short rate $r_t = y_t^0$ is driven by some d-dimensional vector of state variables, X_t, such that $r_t = \delta_0 + \delta_1' X_t$, and the dynamics of X_t are

$$dX_t = \kappa(\theta - X_t)dt + \Sigma\sqrt{S(X_t)}dW_t, \tag{2}$$

where W_t is a d-dimensional Brownian motion under the physical measure $\mathbb{P}$, κ and Σ are $d \times d$ matrices, δ_1 and θ are $d \times 1$ vectors, and δ_0 is a scalar. In addition, $S(X_t)$ is a diagonal $d \times d$ matrix with each element affine in X_t,

$$[S(X_t)]_{ii} = \alpha_i + \beta_i' X_t, \tag{3}$$

with α_i a scalar, and β_i a $d \times 1$ vector. Given a suitable market price of risk specification λ_t (see Appendix C), the dynamics of X_t are governed by an affine diffusion under the risk-neutral measure $\mathbb{Q}$, too,

$$dX_t = \tilde{\kappa}(\tilde{\theta} - X_t)dt + \Sigma\sqrt{S(X_t)}dW_t^{\mathbb{Q}}. \tag{4}$$

Following Duffie and Kan (1996), the yields are given by

$$y_t^{\tau} = \frac{A(\tau)}{\tau} + \frac{B(\tau)'}{\tau} X_t, \tag{5}$$

where A, B are solutions to the system of ordinary differential equations

$$\begin{aligned} \frac{dA(\tau)}{d\tau} &= \tilde{\theta}'\tilde{\kappa}' B(\tau) - \frac{1}{2}\sum_{i=1}^{d}\left[\Sigma' B(\tau)\right]_i^2 \alpha_i + \delta_0, \ A(0) = 0, \\ \frac{dB(\tau)}{d\tau} &= -\tilde{\kappa}' B(\tau) - \frac{1}{2}\sum_{i=1}^{d}\left[\Sigma' B(\tau)\right]_i^2 \beta_i + \delta_1, \ B(0) = 0, \end{aligned} \tag{6}$$

which can be solved either analytically or numerically.

By Eq. (5), the h periods ahead conditional variance of the maturity τ yield, given information through t, is

$$V^{\tau}_{t,h} = Var_t(y^{\tau}_{t+h}) = \frac{1}{\tau^2} B(\tau)' Var_t(X_{t+h}) B(\tau) = b_0^{\tau,h} + b_1^{\tau,h\prime} X_t, \tag{7}$$

i.e., affine in the latent state variables. The precise forms of $b_0^{\tau,h}$ and $b_1^{\tau,h}$ are given in Appendix D.3. We consider affine models with d, the number of factors, from 1 to 3. Table 1 summarizes the models considered, including reference, $A_m(d)$ classification as in Dai and Singleton (2000), m being the number of factors conditioning variances, and market price of risk specification. In Appendix C, we provide a formal description of each model. We use Eq. (7) as the basis of our first yield curve based forecasters. Given the diversity of models considered, we are able to examine the impact of both the number of latent factors and stochastic (state dependent) volatility on the forecasting of future interest rate risk.

Implementation of Eq. (7) requires values for $b_0^{\tau,h}$, $b_1^{\tau,h}$, and X_t. We estimate the models recursively by quasi maximum likelihood (QML) using the Kalman filter and an expanding estimation window, allowing for additive measurement errors in yields (see Section 4 for details). To construct the forecast at t' of subsequent realized yield volatility $V^{\tau}_{t'+1\to t'+h}$, only information through s is used to generate $b_0^{\tau,h}$, $b_1^{\tau,h}$, and $X_1, \dots, X_s$, and hence the conditional volatility (or variance) estimate $\hat{V}^{\tau}_{s,h}$ from Eq. (7), for the given τ and h. This procedure is repeated for $s = 1, \dots, t'$. Next, Eq. (1) is estimated with $Z_t = \hat{V}^{\tau}_{t,h}$, $t = 1, \dots, t' - h$, i.e., $q = 1$ in this case, and the final forecast of $V^{\tau}_{t'+1\to t'+h}$ is $\hat{\rho}_0^{\tau,h} + \hat{\rho}_1^{\tau,h\prime}\hat{V}^{\tau}_{t',h}$. Thus, even with correct model specification, the conditional variance itself is not an unbiased forecast of subsequent realized yield volatility. By the law of total variance, it includes the variance of the conditional mean yield, given both information through t' and integrated volatility

through $t'+h$, thus inducing an upward bias.[3] The recursive regressions in Eq. (1) provide a simple means of converting conditional variances into proper forecasts of future interest rate risk based on investor's information set.

2.1.2. PCA based forecasters

To examine whether or not the yield curve contains relevant information about future volatility and, if it does, whether the term structure models capture this information, we construct volatility forecasts using PCA factors based on the yield curve. We assume that the $N \times T$ yield panel is generated by a factor model,

$$y_t = \pi_0 + \pi_1' F_t + \varepsilon_t, \tag{8}$$

where F_t is a k-vector of common covariance-generating factors, $k < N$, π_1 is a $k \times N$ matrix of factor loadings, and π_0 represents de-meaning. We estimate F_t by PCA using the methodology of Bai and Ng (2002). There is some consensus in the literature that three factors can explain the variation in the yield curve (e.g., Litterman and Scheinkman (1991)). However, whether more factors can improve volatility forecasts, and hence whether different factors are important for explaining yields and for forecasting volatilities are open questions.

Recursive volatility forecasts are built using the fitted PCA factors $\hat{F}_t$ from Eq. (8) for Z_t in Eq. (1), for fixed τ, h, i.e., $q = k$ in this case. In the empirical analysis, we use $k = 3$ to 6 factors.

2.1.3. A common factor approach

For our third yield curve based forecaster, we consider a common factor approach, using volatility information across maturities to combine the maturity-specific forecasts from Section 2.1.2. The approach is similar to that used for forecasting bond risk premiums in Cochrane and Piazzesi (2005), based on forward rates, and in Ludvigson and Ng (2009), based on macro variables. Here, we construct a common factor for interest rate risk forecasting based on information from the yield curve.

Let $\tilde{F}_t$ denote some combination of fitted PCA factors from Eq. (8). We regress subsequent variance averaged across maturities on $\tilde{F}_t$,

$$\frac{1}{4}\sum_{i=1}^{4} V^{\tilde{\tau}_i}_{t+1\to t+h} = \gamma_0^h + \gamma_1^{h\prime}\tilde{F}_t + w^h_{t+h}, \tag{9}$$

for fixed h, where $\tilde{\tau}_1, \ldots, \tilde{\tau}_4$ are the four maturities for which we have high-frequency data. Recursive volatility forecasts are constructed using Eq. (1), for each maturity $\tau = \tilde{\tau}_i$, separately, with the fitted values $\tilde{V}_t$ from Eq. (9) for Z_t, i.e., $q = 1$ in this case. In $\tilde{F}_t$, we allow for selection from among the first six PCA factors from Eq. (8), and in robustness checks we allow squares, cubes, and interactions in the first three of these (see Section 6.1). Following Ludvigson and Ng (2009), the optimal combination of terms to include in $\tilde{F}_t$ is selected by minimizing the Bayesian information criterion (BIC) for Eq. (9). We consider both selection based on the initial estimation period, only, and recursive updating of the selection every period.

2.1.4. Risk premium based forecasters

Motivated by the literature on forecasting risk premiums in the bond market, we use interest rate spreads to form two additional yield curve based volatility forecasters. The first is the yield spread of Campbell and Shiller (1991). In this case, the explanatory variable for the maturity τ yield volatility is simply the difference between the maturity τ yield itself and the short rate, which we proxy by the CRSP 1-month T-bill rate, denoted $y_t^{1/12}$. Thus, volatility forecasts are constructed from Eq. (1), with $Z_t = y_t^\tau - y_t^{1/12}$, i.e., $q = 1$.

The final risk premium based forecaster is based on the forward spread of Fama and Bliss (1987). We use the difference between the maturity τ forward rate, $f_t^\tau = y_t^\tau + (\tau - h)\left(y_t^\tau - y_t^{\tau-h}\right)/h$, and the short rate as explanatory variable. Thus, maturity τ yield volatility forecasts are constructed from Eq. (1), with $Z_t = f_t^\tau - y_t^{1/12}$.

2.2. Time series based yield volatility forecasters

As benchmarks, we consider three simple time series models, namely, the random walk (RW), the HAR model of Corsi (2009), and the realized GARCH model of Hansen et al. (2012). All three are widely used for volatility forecasting in equity markets. The RW forecast of $V^\tau_{t+1\to t+h}$ is simply $V^\tau_{t-h+1\to t}$. The realized GARCH specification is chosen because it utilizes the information in the high-frequency data in forecasting, similarly to RW and HAR.

2.2.1. The HAR model

The HAR model is given by

$$V^\tau_{t+1\to t+h} = \beta_0^{\tau,h} + \beta_D^{\tau,h} V_t^\tau + \beta_W^{\tau,h} V^\tau_{t-4\to t} + \beta_M^{\tau,h} V^\tau_{t-21\to t} + u^{\tau,h}_{t+h}. \tag{10}$$

The motivation for the cascade specification is the hypothesis on heterogeneous beliefs, according to which different investors react to information from the past day, week (5 trading days), and month (22 days). The model is estimated using nonoverlapping dependent variables, i.e., estimation given information through t' uses $t = t'-h, t'-2h, \ldots$ in Eq. (10). Upon estimation, the forecast as of t' is given directly by Eq. (10), for $t = t'$, and $u^{\tau,h}_{t'+h} = 0$.

2.2.2. Mean-reverting realized GARCH

Our final time series benchmark is the realized GARCH model. To accommodate mean reversion in yields, we consider a slightly extended model specification, given by

$$y_t^\tau - y^\tau_{t-h} = a^\tau + b^\tau y^\tau_{t-h} + \sqrt{h_t^\tau} z_t^\tau, \tag{11}$$

$$h_t^\tau = c^\tau + \sum_{i=1}^{p} d_i^\tau h^\tau_{t-ih} + \sum_{j=1}^{q} g_j^\tau V^\tau_{t-(j+1)h+1\to t-jh}, \tag{12}$$

$$V^\tau_{t-h+1\to t} = \xi^\tau + \phi^\tau h_t^\tau + \vartheta_1^\tau z_t^\tau + \vartheta_2^\tau((z_t^\tau)^2 - 1) + u_t^\tau, \tag{13}$$

where h_t^τ is the conditional variance of y_t^τ, given information through $t-h$, $z_t^\tau \sim \mathcal{N}(0,1)$, $u_t^\tau \sim \mathcal{N}(0, \sigma^2_{\tau,u})$, and suppressing the dependence of parameters on h. The standard model for equities, with returns (log-price changes) rather than yield changes on the left-hand side of Eq. (11), corresponds to the special case $b^\tau = 0$, and the extended model accommodates mean reversion in yields for $b^\tau \in (-1, 0)$ (at a rate corresponding to $\kappa = -\log(1 + b^\tau)/h$ in continuous time, see Appendix D). For purposes of forecasting volatility over the $h = 22$ day horizon, we consider a monthly specification, i.e., given information through t', the model is fit to yields and realized measures at $t = t', t'-h, t'-2h, \ldots$, in analogy with the HAR model.

The model is estimated by maximum likelihood. We adopt the specification $p = 1$, $q = 2$ from Hansen et al. (2012). Upon estimation, the forecast as of t' is $\hat{\xi}^\tau + \hat{\phi}^\tau h^\tau_{t'+h}$. For comparison, we applied the standard model with either yields or yield changes on the left-hand side of Eq. (11), i.e., imposing $b^\tau = -1$ and 0, respectively. From the results, the extended mean-reverting realized GARCH model performs best for purposes of yield volatility forecasting, so we focus on this specification in our empirical work.[4]

[3] Write $\mathcal{F}_t$ for the information set. We have $Var(y^\tau_{t+h} \mid \mathcal{F}_t) = \mathbb{E}(Var(y^\tau_{t+h} \mid \mathcal{F}_t, IV^\tau_{t\to t+h}) \mid \mathcal{F}_t) + Var(\mathbb{E}(y^\tau_{t+h} \mid \mathcal{F}_t, IV^\tau_{t\to t+h}) \mid \mathcal{F}_t)$, where $Var(y^\tau_{t+h} \mid \mathcal{F}_t, IV^\tau_{t\to t+h}) = IV^\tau_{t\to t+h}$. Hence, $V^\tau_{t,h} = Var(y^\tau_{t+h} \mid \mathcal{F}_t)$ equals $\mathbb{E}_t(IV^\tau_{t\to t+h}) \approx \mathbb{E}_t(V^\tau_{t+1\to t+h})$ (as realized volatility converges rapidly to integrated volatility in high-frequency data) plus a positive bias stemming from conditional mean variation, $Var(\mathbb{E}(y^\tau_{t+h} \mid \mathcal{F}_t, IV^\tau_{t\to t+h}) \mid \mathcal{F}_t)$. The latter is negligible for short horizons h (cf. Andersen et al. (2006)), but potentially important for our month-ahead forecasts.

[4] Results for the standard realized GARCH model are available from the authors on request.

3. Data description and construction of realized measure

3.1. High-frequency data approach

Our high-frequency data set is based on the three Treasury futures (5 years, 10 years, and long term) for the long end of the yield curve, and 3-month Eurodollar futures for the short end. The Treasury futures are traded on the Chicago Board of Trade (CBOT), and the Eurodollar futures on the Chicago Mercantile Exchange (CME). Data are obtained from CME Group. For liquidity reasons, we start our sample on January 2, 2000, and data run through December 31, 2020. Due to the opening hours in the early part of our sample, we consider a trading day from 7:20 am to 2:00 pm Eastern Time. All weekends and non-working days are excluded. This leaves a raw data set covering 5,115 days.

The Treasury futures have delivery dates in four different months during the year—March, June, September, and December. For each underlying bond maturity, we include the futures contract with highest liquidity, in most cases that with shortest term to delivery. For the Eurodollar futures, we include the two contracts with closest to 3 and 9 months to delivery, and convert the prices of these into yields of 6 months and 1 year to maturity. Section 3.2 describes the details of the procedure.

We extract high-frequency yield curves from the five futures prices in a two-step approach. In the first step, futures prices are converted into coupon bond prices. In the second, yield curves are extracted from the coupon bond prices. Our approach differs from that of Faust et al. (2007) mainly in two respects. First, we include more information on the long end of the curve, using all three Treasury futures, rather than only 10-year contracts. Second, following, e.g., Cieslak and Povala (2016), we calibrate yield curves using the spline method of Fisher et al. (1995), instead of the Nelson and Siegel (1987) method.[5] In the following, we briefly outline the approach.

3.2. Construction of intraday bond prices from futures data

Both Treasury and Eurodollar futures are considered in the first step. The underlying bond of a Treasury futures contract is hypothetical, and unknown until delivery. The seller chooses a bond from a delivery basket consisting of bonds meeting requirements from CME, implying that a Treasury futures involves two options: When to deliver, and which bond to deliver. We neglect the value of the first option by assuming that the delivery date is the first working day of the delivery month. Next, given the delivery date, we construct the delivery basket by combining the futures data with daily CRSP data on Treasuries, find the cheapest-to-deliver (CTD) bond in the basket thus constructed, and calculate the futures-implied coupon bond price.[6] More details on this step are provided in Appendix A.1.

The value of the Eurodollar futures at maturity is determined by the 3-month LIBOR rate. From the observed futures prices, we derive high-frequency LIBOR rates with maturity date 3 months after the delivery date. LIBOR rates are afflicted with credit risk, and we deduce Eurodollar-implied government yields by assuming a constant credit spread through the day between LIBOR and government yields. Daily LIBOR rates with maturities of 6 months and 1 year are obtained from the St. Louis Fed and compared with daily Gürkaynak et al. (2007) yields (see Section 3.5 for the construction) to estimate the spread.

Finally, Eurodollar-implied government yields are converted into zero-coupon bond prices, so that units match with the coupon bonds.

For each of the five contracts (three Treasury and two Eurodollar futures), we construct minute bars containing the first observation in each minute. If multiple observations on the same contract have the same time stamp, we use the median price. We apply a sanity check, discarding observations at a distance more than ten times the absolute mean from the median of the previous ten observations. Finally, we exclude days without any observation for a whole hour for at least one of the five futures contracts. This eliminates 96 trading days, leaving 5,019 observation days in the final analysis data set. Appendix B describes the liquidity of the Treasury contracts.

3.3. Construction of intraday yield curves from bond prices

In the second step, extracting high-frequency yield curves from the futures-implied bond prices using cubic splines, an implication of our approach is that the maturities of bonds considered vary over the sample period, and therefore knot points vary, too. Due to the small number of cross-sectional observations (five futures contracts), we choose not to interpolate the calibrated curves between points of principal payments. The outcome of the procedure is a complete set of consecutive yield curves at the 1-minute frequency. More details on the second step are provided in Appendix A.2.

Yields are read off the calibrated curves at maturities τ reflecting the underlying maturities of the futures contracts, to ensure a sufficient amount of market based information around the relevant points along the curves. Initially, we consider $\tau = 0.5$, 1, 5, 7, and 15 years. The match to the underlying maturities of the corresponding futures is by construction for the first two, and holds to a reasonable degree for the third, while $\tau = 7$ is included as the closest among standard maturities to the approximately 7.5 year average maturity of the Treasury note underlying the 10-year contract. The underlying of the long term Treasury futures is a hypothetical 6% bond. At expiration, a bond with maturity between 15 and 25 years must be delivered. When the interest rate is below 6%, conversion rates favor bonds with short maturities, i.e., the underlying bond is actually of maturity 15 years for our out-of-sample window, and hence $\tau = 15$ is considered as a candidate fifth maturity for our construction. To verify the quality of the resulting high-frequency yields at the five candidate τ-values, we match a daily frequency subsample of them with daily Gürkaynak et al. (2007) yields. For the latter, the number of bonds with maturity around 15 years outstanding at any given point in time in the CRSP data is rather limited, hence implying a low weight around this maturity in the calibration, and we find a relatively low correlation between the resulting yields and the daily subsample of ours at $\tau = 15$. Thus, as we cannot verify the quality of our fit at the fifth maturity, we henceforth restrict attention to the yields read off our high-frequency curves at $\tau = 0.5$, 1, 5, and 7 years, although we continue to use all five futures contracts in the construction of the curves.

At the four maturities retained, correlations between the end-of-day subset of futures-implied yields extracted from our 1-minute data and the daily calibrated yields range between 99.37% and 99.99% (see Table A.1 in the Appendix). In effect, we have created a high-frequency intraday version of the Gürkaynak et al. (2007) dataset up to maturity $\tau = 7$. Table 2 shows descriptive statistics on our yield data, by maturity. On average over the sample period, the term structure of interest rates is mainly upward sloping, and the term structure of volatilities downward sloping, although both are relatively flat at the short end. Skewness and kurtosis are modest, especially at longer maturities. Fig. 1 displays the evolution through time in yields, by maturity, along with a three-dimensional view of the evolution of the high-frequency yield curves through calendar time. The transition to the zero lower bound (ZLB) regime around 2008 has a strong impact, especially at the shorter maturities.

[5] A related cubic spline method due to Waggoner (1997) is adopted by Dai et al. (2007) and Andersen and Benzoni (2010).

[6] Treasuries are non-convertible, and starting in 2000 avoids problems with the flower bonds issued until 1965, as these all matured by 1998. Callable bonds and notes were issued until 1985, but many of these subsequently repurchased by the Treasury and reissued as non-callable, although on a discretionary basis, without sinking fund provision. We drop the small number of remaining callable issues outstanding as of January 2, 2000.

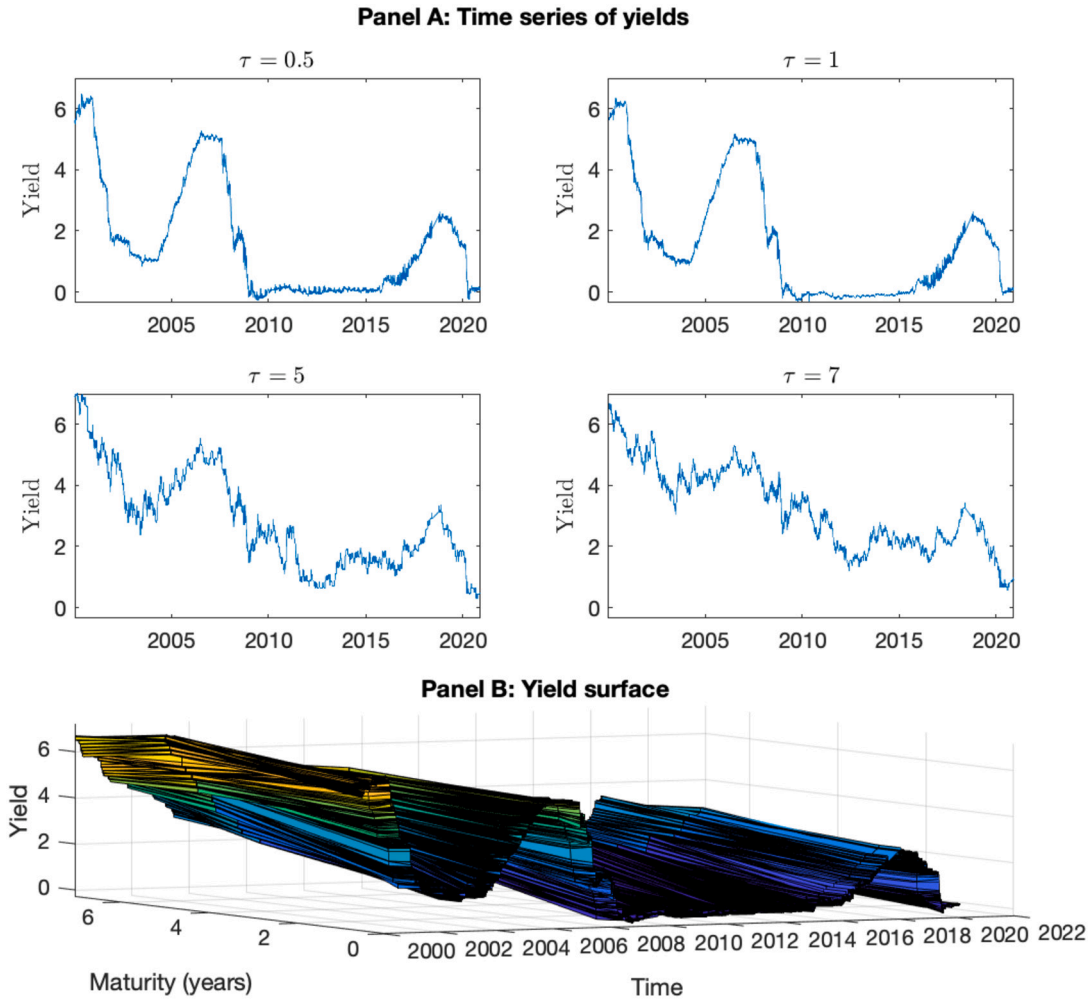


Fig. 1. High-frequency yields.
This figure shows the time series of high-frequency annualized yields in percent, by maturity, along with a three-dimensional view of the evolution through time of the yield curves. The sample spans the period from January 2, 2000, through December 31, 2020.

Table 2
Descriptive statistics on high-frequency yields.

	$\tau = 0.5$	$\tau = 1$	$\tau = 5$	$\tau = 7$
Mean	1.63	1.57	2.82	3.25
Std.	1.86	1.88	1.59	1.35
Skewness	1.06	1.01	0.60	0.26
Kurtosis	2.94	2.84	2.50	2.19

This table presents the mean, standard deviation, skewness, and kurtosis for the end-of-day subset of our high-frequency annualized yields in percent, by maturity. The sample spans the period from January 2, 2000, through December 31, 2020.

3.4. Realized volatility measure

Access to high-frequency data on intraday yield curves enables us to estimate volatilities using realized measures. The market for the Treasury futures used in the construction of yield curves (Section 3.2) has undergone dramatic changes in liquidity during our sample period, in part due to the introduction of the electronic trading pit in 2004, and data are likely afflicted with market microstructure noise (see Appendix B). The high frequency econometrics literature proposes estimators of volatility that are robust to microstructure noise, e.g., the two time scales estimator of Zhang et al. (2005), the realized kernel approach of Barndorff-Nielsen et al. (2008), and the pre-averaged realized variance of Jacod et al. (2009). We deal with the issue by implementing the pre-averaging estimator following Christensen et al. (2014), who show that this is successful in empirical analysis of volatility.

For each maturity τ, and on each day, write the intraday yield data as $y^{\tau}_{\frac{i}{n}}$, for $i = 0, 1, \ldots, n$. Pre-averaged yield changes are calculated by averaging yields in a local neighborhood consisting of K observations,

$$\Delta\bar{y}^{\tau}_{i} = \frac{1}{K}\left(\sum_{j=K/2}^{K-1} y^{\tau}_{\frac{i+j}{n}} - \sum_{j=0}^{K/2-1} y^{\tau}_{\frac{i+j}{n}}\right), \tag{14}$$

with K the nearest even integer to $\theta\sqrt{n}$, and θ a tuning parameter. The pre-averaged realized variance is then calculated as

$$V^{\tau}_{t} = \frac{n}{n-K+2}\frac{1}{K\psi_K}\sum_{i=0}^{n-K+1}\left(\Delta\bar{y}^{\tau}_{i}\right)^2 - \frac{\hat{\omega}^2}{\theta\psi_K}, \tag{15}$$

with $\psi_K = (1 + 2K^{-2})/12$, and $\hat{\omega}^2$ an estimate of the noise variance,

$$\hat{\omega}^2 = -\frac{1}{n-1}\sum_{i=2}^{n}\left(y^{\tau}_{\frac{i}{n}} - y^{\tau}_{\frac{i-1}{n}}\right)\left(y^{\tau}_{\frac{i-1}{n}} - y^{\tau}_{\frac{i-2}{n}}\right). \tag{16}$$

Given our one-minute sampling, the number of observations in a day, n, is 380, and we set $\theta = 1$. Since we are interested in volatility over the entire day, not only the hours for which we have high-frequency data, we add the squared overnight difference in yields, following Andersen

Table 3
Descriptive statistics on pre-averaged realized yield volatility.

	$\tau=0.5$	$\tau=1$	$\tau=5$	$\tau=7$
Panel A: One-day volatility, $\sqrt{V_t^\tau}$				
Mean	0.89	0.79	1.28	1.22
Std.	1.05	1.01	1.86	1.51
Skewness	4.33	4.79	2.13	1.78
Kurtosis	45.48	53.80	7.19	6.63
Panel B: Month-ahead volatility, $\sqrt{V_{t+1\|t+22}^\tau}$				
Mean	1.16	1.01	2.14	1.87
Std.	0.81	0.84	0.92	0.69
Skewness	2.41	2.04	-0.09	0.60
Kurtosis	11.57	9.08	2.54	3.68
Panel C: One-day variance, V_t^τ				
Mean	1.88	1.64	5.12	3.75
Std.	8.62	8.48	12.98	8.50
Skewness	31.07	33.72	3.84	6.17
Kurtosis	1462.90	1663.60	21.09	101.05
Panel D: Month-ahead variance, $V_{t+1\|t+22}^\tau$				
Mean	1.99	1.73	5.42	3.96
Std.	3.74	3.43	4.00	2.97
Skewness	5.49	5.24	0.89	1.83
Kurtosis	41.24	39.13	3.58	7.98

This table presents the mean, standard deviation, skewness, and kurtosis of the daily annualized pre-averaged realized yield volatilities, by maturity. Statistics for the square root (volatility) form are shown for one-day measures in Panel A, and for one-month ($h=22$) measures in Panel B, both in percent. Statistics for the raw (variance) form are shown for one-day measures in Panel C, and for one-month ($h=22$) measures in Panel D, both in basis points. The sample spans the period from January 2, 2000, through December 31, 2020.

and Benzoni (2010) and Bollerslev et al. (2018). The risk measure over the next month is simply $V_{t+1\to t+h}^\tau$, $h=22$, aggregating the pre-averaged realized variances over 22 trading days.

Table 3 shows descriptive statistics on the daily annualized pre-averaged realized yield volatilities over one day, $\sqrt{V_t^\tau}$, in Panel A, and one month, $\sqrt{V_{t+1\to t+22}^\tau}$, in Panel B, in percent, by maturity. Although the volatility (square root) form of the measures facilitates interpretation, e.g., a unit mean corresponds to a one percent annual yield volatility, the raw (variance) form is used for forecasting, and descriptive statistics for this are reported in Panels C and D.

The term structure of volatilities (variances) exhibits a hump shape across the four maturities considered, with highest average at the $\tau=5$ year maturity. The standard deviation (time series variation in volatility) shows a similar pattern. Skewness and excess kurtosis are higher for shorter maturities, $\tau=0.5$ and 1, relative to longer, $\tau=5$ and 7, where they essentially vanish for the one-month volatility measure.[7] On average, volatilities over one month are slightly larger than over one day, possibly reflecting that some conditional mean variation remains in the realized measures due to finite sampling frequency, cf. footnote 3, as well as Jensen's inequality for the square root measures (volatility spikes are included in 22 measures rather than one). As expected, other moments are smallest for the aggregated measures.

Fig. 2 displays the evolution through time in daily annualized pre-averaged realized one-month yield volatility in percent, by maturity, corresponding to Table 3, Panel B, along with a three-dimensional view of the volatility surface.

Volatilities rise quite dramatically for the shortest maturities during the financial crisis and the transition to the ZLB regime, and drop to a low level after the transition, only to rise again around 2016.

3.5. Daily yield data

To enable estimation of the affine term structure models, as well as the PCA, common factor, and risk premium based forecasters, on a daily yield panel with more than four observations in the cross section,[8] we consider the estimated parameters provided by Gürkaynak et al. (2007). Their daily frequency dataset is extracted from a large set of coupon bonds, using the Svensson (1994) method. The continuously compounded yield at maturity τ is written as

$$y_t^\tau=\beta_0+\beta_1\frac{1-e^{-\frac{\tau}{\theta_1}}}{\frac{\tau}{\theta_1}}+\beta_2\left(\frac{1-e^{-\frac{\tau}{\theta_1}}}{\frac{\tau}{\theta_1}}-e^{-\frac{\tau}{\theta_1}}\right)+\beta_3\left(\frac{1-e^{-\frac{\tau}{\theta_2}}}{\frac{\tau}{\theta_2}}-e^{-\frac{\tau}{\theta_2}}\right), \quad (17)$$

and estimates of $(\beta_0,\beta_1,\beta_2,\beta_3,\theta_1,\theta_2)$ are provided at daily frequency.[9] We follow Christensen et al. (2010) and consider $N=8$ maturities in the daily cross sections, namely, 3 and 6 months, and 1, 2, 3, 5, 7, and 10 years.

4. Estimation

The affine term structure models from Section 2.1.1 are estimated on daily data, cf. Section 3.5, using the Kalman filter. The basic measurement and transition equations are obtained by allowing for measurement error ε_{t+h}^τ in the yields in Eq. (5), and discretizing the state dynamics in Eq. (2), i.e., the state space model is given by

$$y_{t+h}^\tau=\frac{A(\tau)}{\tau}+\frac{B(\tau)'}{\tau}X_{t+h}+\varepsilon_{t+h}^\tau, \quad (18)$$

$$X_{t+h}=C_h+D_h'X_t+\eta_{t+h}, \quad (19)$$

where $\varepsilon_{t+h}^\tau\sim\mathcal{N}(0,H_{\tau,h})$, $\eta_{t+h}\sim\mathcal{N}(0,Q_{t,h})$, $h=1$ for daily data and daily time index t, with expressions for A and B in Appendix C, and for C_h, D_h, and $Q_{t,h}$ in Appendix D. For the two Gaussian models in Table 1, the standard linear filter applies. For the stochastic volatility models, Cox et al. (1985) (henceforth CIR) and AFNS$_3$, we apply the extended Kalman filter, approximating transitions by Gaussian distributions.[10]

Upon estimation, conditional variance estimates are computed using Eq. (7), now with h indicating the forecasting horizon ($h=22$ for month-ahead forecasting), and corrected for the bias stemming from measurement error in yields, ε_{t+h}^τ in Eq. (18), producing $\tilde{V}_{t,h}^\tau=b_0^{\tau,h}+b_1^{\tau,h\prime}X_t+H_{\tau,h}$. Recursive volatility forecasts are constructed using $\tilde{V}_{t,h}^\tau$ for Z_t in Eq. (1).

5. Empirical results

We consider the forecasting of yield volatility over the next month. The first set of estimates is based on the period January 2, 2000, through December 31, 2007, and the next 100 observations are used

[7] Skewness and kurtosis are close to 0 and 3.

[8] Our high-frequency yield curves are constructed using five futures contacts, and four yields are retained in the resulting high-frequency panel, cf. Section 3.3. The time series models are estimated using realized measures (Section 3.4) based on the high-frequency panel.

[9] Available at http://www.federalreserve.gov/pubs/feds/2006/200628/200628abs.html and updated daily.

[10] Appendix D provides further details on estimation, including the Kalman filter and the (quasi) log likelihood function.

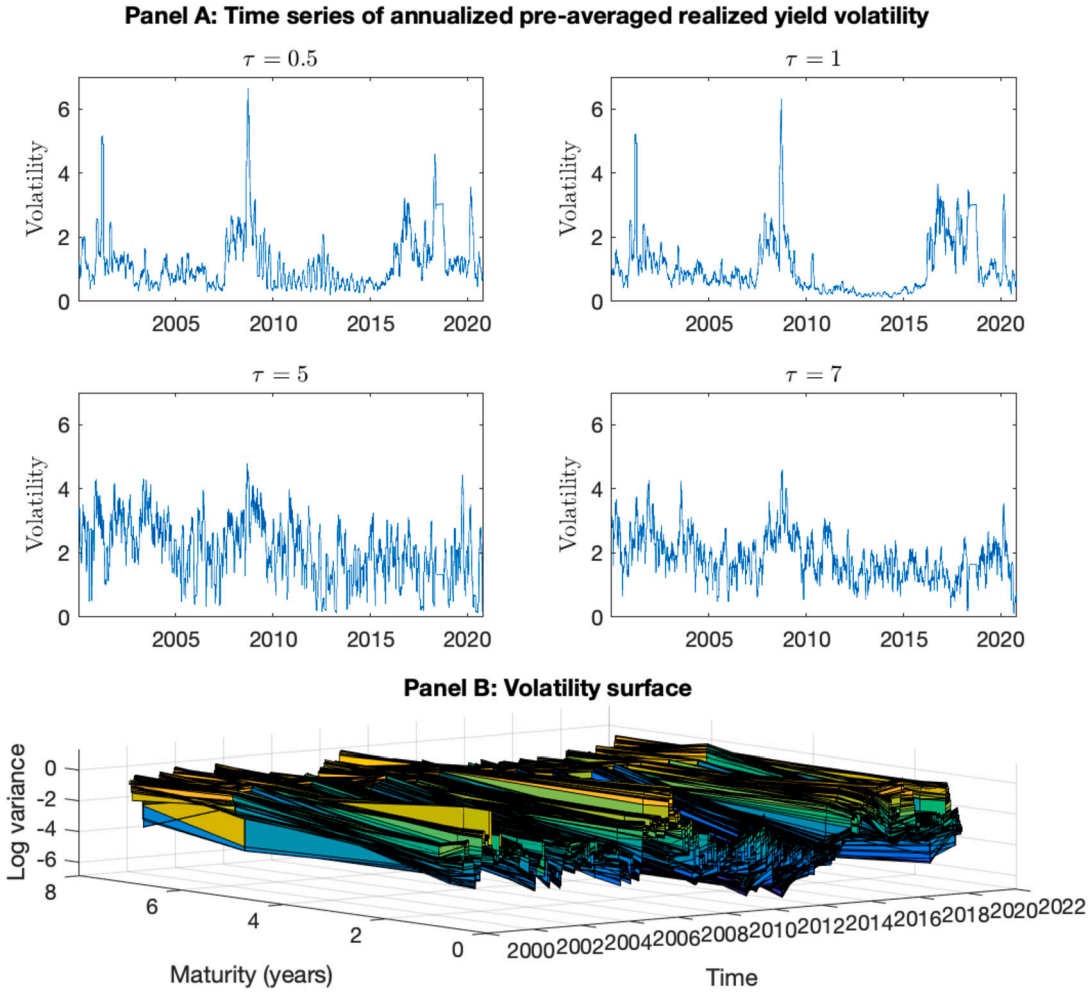


Fig. 2. Annualized pre-averaged realized yield volatility.
This figure shows the time series of daily annualized pre-averaged realized one-month yield volatilities in percent, by maturity, along with a three-dimensional view of the evolution through time of the term structure of volatilities (log variances). The sample spans the period from January 2, 2000, through December 31, 2020.

for recursive estimation of the predictive regressions in Eq. (1). This leaves 2,976 observations for the out-of-sample period, covering June 7, 2008, through December 31, 2020. This way, the out-of-sample window starts just before the transition to the ZLB regime. For PCA, risk premium based forecasters, and the HAR model, forecasts are not restricted to be positive. To ensure meaningful volatility forecasts, we apply a sanity filter, such that we do not forecast below the 2.5 or above the 97.5 percentiles of the empirical distribution of observed realized variances.[11]

5.1. Statistical value of interest rate risk forecasts

To assess the yield volatility forecasts against a RW, we consider the R^2_{OoS} measure of Campbell and Thompson (2007),

$$R^2_{OoS} = 1 - \frac{\sum_{t=t_0+1}^{T-h}(V^{\tau}_{t+1\to t+h} - \hat{V}^{\tau}_{t,h,\xi})^2}{\sum_{t=t_0+1}^{T-h}(V^{\tau}_{t+1\to t+h} - \hat{V}^{\tau}_{t,h,RW})^2}, \tag{20}$$

where $\hat{V}^{\tau}_{t,h,\xi}$ is the forecast from model ξ, t_0 is the end of the initial estimation period, and T is the end of the sample. A positive R^2_{OoS} indicates more accurate volatility forecasts from model ξ than from the RW. The null hypothesis $R^2_{OoS} \leq 0$ is tested against the alternative $R^2_{OoS} > 0$ using a one-sided Diebold and Mariano (1995) test based on the Newey and West (1994) variance estimator with automatic lag selection of Andrews (1991).

Table 4 presents the resulting R^2_{OoS} statistics in percent and Diebold-Mariano p-values for all forecasters and maturities considered. Only the HAR model generates positive R^2_{OoS} statistics across all maturities. Furthermore, for each maturity, HAR generates the highest statistic (most accurate forecasts) across all models, except that the forward spread and the best common factor based forecaster get even higher statistics at $\tau = 5$. From the results, yield volatility forecasting is hardest at the short and long ends of the curve, with only HAR generating positive R^2_{OoS} at $\tau = 0.5$, 1, and 7. In contrast, at the intermediate maturity $\tau = 5$, R^2_{OoS} is positive for all models, except Vasicek.

A few comparisons within the first panel of Table 4, the results for the term structure models, are worth noting. Among the four models considered, Vasicek shows the strongest forecasting performance at the two shortest maturities, and the weakest at the two longest. Introducing stochastic volatility only leads to improved forecasts in half of the cases considered, i.e., at the two longest maturities for the one-factor model, and at the two shortest for the three-factor model.

Although constant-volatility models are typically rejected in-sample, parsimony can be rewarded in out-of-sample forecasting comparisons, as it reduces parameter uncertainty and the risk of overfitting, albeit at the expense of increased risk of model misspecification. In the present case, the results suggest that parsimony is rewarded at shorter maturities, with the single-factor homoskedastic model performing best, and

[11] The empirical distribution of observed variances is updated recursively, and forecasts below the 2.5 and above the 97.5 percentiles replaced by these.

Table 4
Out-of-sample R^2 for month-ahead yield volatility forecasting.

	$\tau = 0.5$	$\tau = 1$	$\tau = 5$	$\tau = 7$
Term structure models				
Vasicek	-15.67	-6.90	-0.84	-52.48
	(0.88)	(0.70)	(0.53)	(1.00)
CIR	- 28.90	-18.55	11.91	-9.55
	(0.96)	(0.87)	(0.19)	(0.70)
$AFNS_0$	-52.53	-48.55	18.95	-14.20
	(1.00)	(1.00)	(0.06)	(0.77)
$AFNS_3$	-40.94	-28.43	4.69	-49.04
	(0.99)	(0.91)	(0.37)	(0.99)
PCA based forecasters				
PCA_3	-32.11	-24.35	15.92	-15.84
	(0.92)	(0.91)	(0.04)	(0.91)
PCA_4	-32.28	-24.48	15.04	-16.67
	(0.93)	(0.92)	(0.06)	(0.92)
PCA_5	-32.48	-23.68	12.72	-23.82
	(0.93)	(0.91)	(0.12)	(0.94)
PCA_6	-39.95	-30.39	10.84	-20.88
	(0.97)	(0.98)	(0.16)	(0.85)
Risk premium based forecasters				
Forward spreads	-28.51	-22.41	22.00	-12.92
	(0.88)	(0.94)	(0.01)	(0.84)
Yield spreads	-28.32	-21.57	8.84	-14.50
	(0.87)	(0.93)	(0.26)	(0.90)
Common factor based forecasters				
Initial	-27.66	-21.72	20.62	-2.25
	(0.88)	(0.90)	(0.02)	(0.59)
Recursive	-38.11	-34.45	17.96	-7.44
	(0.95)	(0.98)	(0.06)	(0.68)
Time series models				
HAR	1.03	7.54	20.44	10.70
	(0.47)	(0.29)	(0.00)	(0.11)
Mean-reverting realized GARCH	-26.83	-20.50	0.38	-38.20
	(0.86)	(0.93)	(0.49)	(1.00)

This table displays R^2_{OoS} measures in percent relative to a RW for all forecasting methods and maturities. In parentheses asymptotic p-values for a one-sided Diebold-Mariano test using the Newey-West variance estimator with automatic lag selection of Andrews (1991). For common factor based forecasters, the label "Initial" indicates that the selection of PCA factors is based on the initial estimation period, and "Recursive" that it is updated every period. The initial estimation period ranges from January 2, 2000, through June 6, 2008, and the out-of-sample period from June 7, 2008, through December 31, 2020.

to some extent at longer maturities, too, with rewards to using either a single factor (CIR), or constant volatility ($AFNS_0$). Further, the CIR model strikes a better balance between parsimony and misspecification than $AFNS_3$, as it dominates this across all maturities.

Based on the results from the term structure models, the general economic message is that complex time-varying models are hard to estimate, to the point that assuming and updating a simple, possibly misspecified model can work better in practice, for purposes of forecasting yield volatility. The similar phenomenon is known from the options literature, e.g., Dumas et al. (1998) find that while state-dependent volatility models improve on the Black and Scholes (1973) model in-sample, updating the constant-volatility model is preferred for out-of-sample hedging.[12] In the fixed income case, updating is in the spirit of the Heath et al. (1992) approach of conditioning on current information, and Buraschi and Corielli (2005) find that it partly corrects for model misspecification. Our results are consistent with these findings.

The next panel in Table 4 shows results for the PCA based forecasters. Including more than three components actually makes forecasts deteriorate, presumably due to loss of parsimony. This is consistent with Litterman and Scheinkman (1991), who found that three factors provide a good description of the term structure. By a rotation argument, one might expect that the three factors in $AFNS_0$ correspond closely to the three leading PCA factors, so the difference in performance is perhaps surprising, with the PCA method generating more accurate forecasts at shorter maturities, and $AFNS_0$ at longer. However, unlike the PCA based volatility forecast, the affine model variance forecast does not involve the latent factors in the Gaussian case, as we show in the Appendix.[13] Presumably, this explains the difference in performance. Consistently with the findings from the term structure models (first panel), the results for the three-factor case indicate rewards to parsimony (updating of the constant-volatility model) at longer maturities, and to the use of time-varying factors ($AFNS_3$, or PCA) at shorter.

For the common factor based forecasters in Table 4, fourth panel, the label "Initial" indicates that the selection of PCA factors in Eq. (9) is based on the initial estimation period, only, and "Recursive" that it is updated every period. From the results, selection based on the initial estimation period generates more accurate forecasts than recursive updating. This finding points to a regime switch in which the ZLB regime is more similar to the initial estimation period than to the transition period, with forecasts into the ZLB period deteriorating if the selection is updated during the transition. Further, at each maturity, the common factor based forecaster with initial selection is more accurate than the maturity-specific (i.e., not using a common factor based on Eq. (9)) PCA based forecasters from the second panel. This indicates that volatilities at different maturities are driven by a common factor, and that averaging out maturity-specific noise improves forecasting accuracy. In fact, the common factor based forecaster dominates the term structure models (first panel) at the two longest maturities, as does the forward spread based forecaster (third panel) at $\tau = 5$, suggesting that the yield curve contains predictive information about future interest rate risk beyond that captured by the affine models.

The last panel in Table 4 shows results for the time series models. Across all maturities, the forecasting performance of the mean-reverting realized GARCH model falls short of HAR. Nevertheless, both models dominate the PCA, risk premium, and common factor approaches at the two shortest maturities. The HAR model gets a p-value of 11% at the longest maturity, and below 1% at $\tau = 5$ years, meaning that it significantly outperforms the RW at level 1% at this intermediate maturity. So does the forward spread based forecast at 1%, the PCA based forecast with three components and the common factor based forecast without updating at 5%, and $AFNS_0$ and the recursive common factor based forecast at 10%. All PCA based forecasts (three through six components) get p-values below 20% at $\tau = 5$. The results point to difficulties for any model to reliably outperform the RW at short and long maturities, but also to scope for accurate volatility forecasting across a variety of different methods at intermediate maturities.

For comparison, Table A.3 in the Appendix shows results for volatility forecasts two months ahead ($h = 44$), laid out in the same manner as Table 4. Although significance is lost in most cases, presumably due to higher noise-to-signal ratio at the longer forecasting horizon, the overall pattern is confirmed. For each maturity, HAR generates the highest

[12] Similarly, Bakshi et al. (1997) find that the constant-volatility model performs no worse than stochastic volatility extensions in terms of hedging across most moneyness categories.

[13] For $m = 0$, we have $b_1^{\tau,h} = 0$ in Eq. (7), see Eq. (D.28) in Appendix D.3.

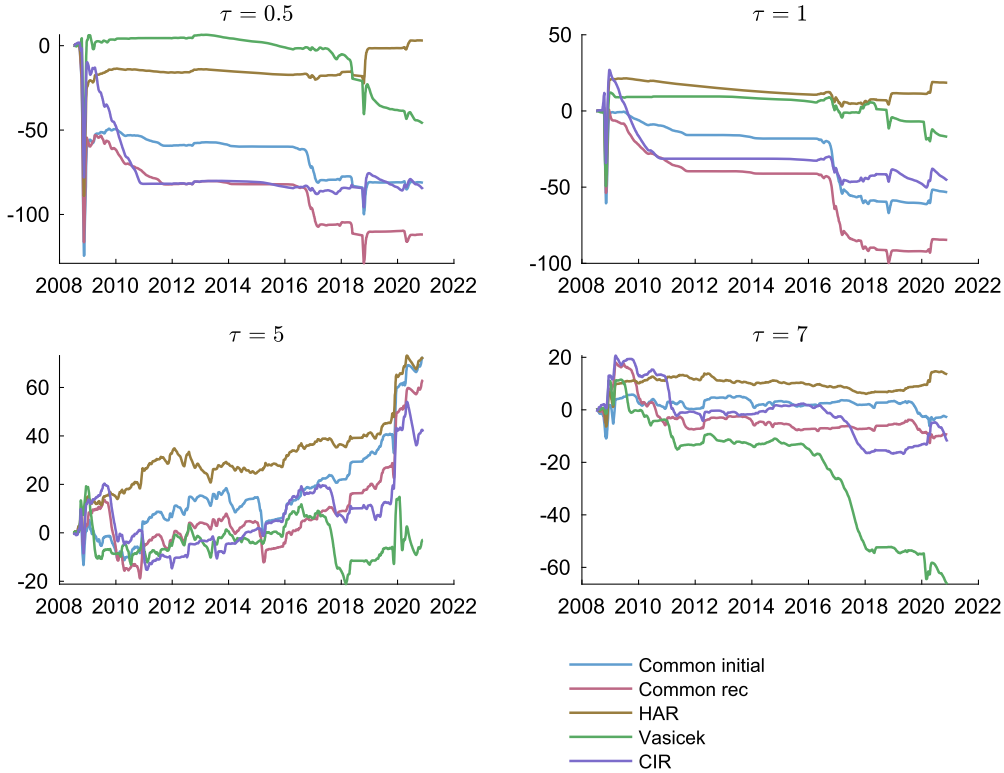


Fig. 3. Yield volatility forecast errors relative to RW.
This figure shows yield volatility forecast errors over time, by maturity, for selected forecasting methods. Cumulative squared errors for a given forecasting method are subtracted from those for the RW, so that an increasing curve indicates better forecasting than by RW, and vice versa. The sample spans the period from January 2, 2000, through December 31, 2020.

R^2_{OoS} statistic (most accurate forecasts) across all models considered, and at $\tau = 5$, the forward spread based forecaster and AFNS_0 get close. Introducing stochastic volatility leads to improved forecasts at $\tau = 5$ for the one-factor model, and at the two shortest maturities for the three-factor model. Although AFNS_3 now performs better than CIR at three maturities, Vasicek continues to dominate the other term structure models at the two shortest. Including more than three components in the maturity-specific PCA based forecaster makes forecasts deteriorate, and initial selection of factors generally generates more accurate forecasts than recursive updating for the common factor approach. The cross section based (PCA, risk premium, common factor) forecasters generally perform better than the term structure models, hence pointing to predictive information in the curve not captured by the affine models. One difference in results is that significance relative to RW is now attained at the shortest maturity for HAR, at level 10%. Henceforth, we focus on the one-month forecasting horizon ($h = 22$).

Fig. 3 shows yield volatility forecast errors over time, by maturity, for selected forecasting methods. Cumulative squared errors for a given forecasting method are subtracted from those for the RW, so that an increasing curve indicates better forecasting than by RW, and vice versa. In the common factor case, the improvement in the approach with initial selection relative to recursive updating occurs around 2009, consistent with the ZLB transition driving the phenomenon. For comparison, the figure shows results for Vasicek, CIR, and HAR, too. Most methods exhibit jumps in forecasting accuracy, moving into the ZLB regime, although the effect materializes more gradually at longer maturities. It could be expected that CIR (with state-dependent volatility) would handle the entire ZLB period better than Vasicek, but this is not confirmed. At the shorter maturities, Vasicek quickly adjusts to the low-volatility regime, whereas CIR performs worse than the RW for about two years. The results show that simpler is better around the transition to ZLB (initial selection as opposed to recursive, constant-volatility model as opposed to CIR), presumably in part because the artificial investor is not ignoring recent volatilities, cf. Eq. (1). Historical volatility (HAR) forecasts better than the curve-based methods, except that Vasicek initially does better at the short end. Improvements relative to the RW accumulate over calendar time at $\tau = 5$, except for the Vasicek model. At other maturities, results stabilize over the ZLB period, which runs through 2016 (cf. Fig. 1), well after the end of quantitative easing by 2014, then deteriorate for most models, HAR being the exception.

Overall, the results show that information from either the yield curve or the time series of volatilities can be used to improve volatility forecasts over the RW. Furthermore, the results indicate that the yield curve contains information about future volatility that is not captured by the term structure models.

5.2. Extracting incremental information

Here, we examine the possibility that the yield volatility forecasting equations based on the affine term structure models suffer from omitted variable bias, i.e., that the expression for future volatility in Eq. (7) should be expanded to

$$V^{\tau}_{t,h} = b^{\tau,h}_0 + b^{\tau,h\prime}_1 X_t + b^{\tau,h\prime}_2 \tilde{X}_t\,, \tag{21}$$

for forecasting purposes. We consider three ways of extracting the incremental information variable $\tilde{X}_t$:
(i) By PCA factors from the yield curve.
(ii) Combining (i) with a factor extracted from the past volatility forecast errors.
(iii) By the realized volatility measure based on high-frequency data.
Since PCA factors capture the shape of the yield curve nonparametrically, improved forecasting performance (here, significance of $b^{\tau,h}_2$) in case (i) indicates that the yield curve contains incremental information about future interest rate risk relative to that in the affine term structure model considered. In (ii), significance indicates the existence of a factor explaining yield volatilities, but not yields, i.e., a USV case. In

Table 5
The incremental information on future volatility in the yield curve.

	$\tau=0.5$	$\tau=1$	$\tau=5$	$\tau=7$
Term structure models				
Vasicek	7.31	1.14	6.39	18.20
	(0.06)	(0.77)	(0.09)	(0.00)
CIR	2.92	4.93	11.85	18.34
	(0.40)	(0.18)	(0.01)	(0.00)
$AFNS_0$	31.18	17.14	0.75	8.49
	(0.00)	(0.00)	(0.86)	(0.04)
$AFNS_3$	4.16	6.97	24.56	82.27
	(0.24)	(0.07)	(0.00)	(0.00)
Time series models				
HAR	5.08	5.41	4.48	18.57
	(0.17)	(0.14)	(0.21)	(0.00)
Mean-reverting realized GARCH	7.81	8.27	6.60	23.73
	(0.05)	(0.04)	(0.09)	(0.00)

This table shows results from regression Eq. (22), explaining yield volatility forecast errors from the specified models using three PCA factors fitted to past yield curves, based on Eq. (8). Reported values are F-statistics for joint tests of $\phi_1^{\tau,h}=0$ (three coefficients). Asymptotic p-values in parentheses. The initial estimation period ranges from January 2, 2000, through June 6, 2008, and the out-of-sample period from June 7, 2008, through December 31, 2020.

Table 6
Specification test for unspanned stochastic volatility factor.

	$\tau=0.5$	$\tau=1$	$\tau=5$	$\tau=7$
Term structure models				
Vasicek	-3.01	-1.65	-1.90	-2.55
	(0.00)	(0.10)	(0.06)	(0.01)
CIR	-9.08	-4.49	-2.43	-3.36
	(0.00)	(0.00)	(0.02)	(0.00)
$AFNS_0$	-5.85	-2.91	-1.12	-2.31
	(0.00)	(0.00)	(0.26)	(0.02)
$AFNS_3$	-8.18	-4.23	-2.09	-1.57
	(0.00)	(0.00)	(0.04)	(0.12)
Time series models				
HAR	-1.35	-0.79	2.48	2.49
	(0.18)	(0.43)	(0.01)	(0.01)
Mean-reverting realized GARCH	-5.54	-2.52	-1.43	-2.30
	(0.00)	(0.01)	(0.15)	(0.02)

This table shows results from regression Eq. (23), explaining yield volatility forecast errors from the specified models augmented with three yield curve PCA factors as in Eq. (22) using a PCA factor fitted to past yield volatility forecast errors. Reported values are t-statistics for testing $\psi_1^{\tau,h}=0$. Asymptotic p-values in parentheses, based on the Newey-West variance estimator with automatic lag selection of Andrews (1991). The initial estimation period ranges from January 2, 2000, through June 6, 2008, and the out-of-sample period from June 7, 2008, through December 31, 2020.

other words, significance of $b_2^{\tau,h}$ shows in case (i) that the term structure models do not capture all relevant information about future volatility available from the yield curve, and in case (ii) that the yield curve itself does not capture all relevant information. In (iii), significance indicates that historical volatility contains incremental information not captured by the term structure models.

In addition, we subject the time series models to the same tests, i.e., we examine whether the similar information variable $\tilde{X}_t$ can be used to improve the volatility forecasts from the time series models. If so, this indicates in (i) that the yield curve contains incremental information about future volatility, relative to that contained in the volatility history itself, and in (iii) that the time series models do not fully exploit the information in the high-frequency data.

5.2.1. The incremental information in the yield curve

For (i), we regress volatility forecast errors from model ξ on PCA factors from the yield curve,

$$V^{\tau}_{t+1\to t+h}-\hat{V}^{\tau}_{t,h,\xi}=\phi_0^{\tau,h}+\phi_1^{\tau,h\prime}\hat{F}_t+u^{\tau,h}_{t+h}, \tag{22}$$

for fixed τ, h, with $\hat{F}_t$ the three leading fitted PCA factors at time t from Eq. (8), based on data through t, and test for joint significance of $\phi_1^{\tau,h}$ ($k=3$ coefficients). Under the null, the model generating the forecast $\hat{V}^{\tau}_{t,h,\xi}$ subsumes the information content on future volatility available in the PCA yield curve factors.

Table 5 presents results from the specification test. In general, the term structure models do not capture all relevant information in the PCA yield curve factors about future volatility. The null is rejected at level 10% or better at two or more maturities for each term structure model, and for two or more models at each maturity. It is rejected at 5% or better for all models at the longest maturity, and at 10% or better at $\tau=5$, except that here, $AFNS_0$ captures the information in the curve about future volatility (p-value 0.86), consistent with the relatively strong performance of this model at the intermediate maturity in Table 4. However, the PCA factors carry incremental information relative to $AFNS_0$ at level 5% for all other maturities. Thus, the indication is that the yield curve contains incremental information about future volatility beyond that captured by the affine models, especially at the long end.

For the time series models, the evidence suggests that the yield curve contains incremental information about future volatility at the long end, relative to that contained in the historical volatility series. The null is rejected at 1% for $\tau=7$ for both the HAR and realized GARCH models, and for the latter at 10% for the other maturities, too.

Taken together, the results indicate that the yield curve contains important incremental information about future volatility relative to the time series of historical volatilities at the long end, and that some of this information is not captured by standard affine models.

5.2.2. Can past forecast errors predict future forecast errors?

In (ii), we examine whether a factor extracted from past yield volatility forecast errors can predict future forecast errors. An implication of USV is that it should be possible to extract at least one factor from volatility which is not related to the yields. We investigate this possibility by recursively extracting a factor, say, $PC^h_{t,\xi}$, from the cross section of current and past forecast errors $V^{\tau'}_{t'+1\to t'+h}-\hat{V}^{\tau'}_{t',h,\xi}$ by applying PCA to forecast errors at $t'=t-h$ and earlier, across maturities τ', using PCA, then testing for whether the extracted factor contains significant information about future forecast errors in the regression

$$V^{\tau}_{t+1\to t+h}-\hat{V}^{\tau}_{t,h,\xi}=\phi_0^{\tau,h}+\phi_1^{\tau,h\prime}\hat{F}_t+\psi_1^{\tau,h}PC^h_{t,\xi}+u^{\tau,h}_{t+h}, \tag{23}$$

for fixed τ, h. Only one factor is included, due to the small number of volatilities in the cross section. The first 100 observations are used to initialize the factor. A significant coefficient $\psi_1^{\tau,h}$ indicates that a serially dependent USV factor is relevant for the forecast.

Table 6 shows results from estimation of Eq. (23). At 5%, evidence of an omitted factor arises at two or more maturities for each term structure model, and for two or more models at each maturity. The coefficients $\psi_1^{\tau,h}$ are negative in all cases, consistent with mean-reversion in the USV factor. The test is conservative, in that the specification in Eq. (23) controls for PCA factors fitted to past yield curves, so the results are indicative of the existence of a latent volatility factor which is

Table 7
The incremental information on future yield volatility in historical volatility.

	$\tau=0.5$	$\tau=1$	$\tau=5$	$\tau=7$
Term structure models				
Vasicek	4.76	7.81	3.19	5.89
	(0.00)	(0.00)	(0.00)	(0.00)
CIR	3.22	4.18	1.57	2.13
	(0.00)	(0.00)	(0.12)	(0.03)
$AFNS_0$	2.15	2.22	1.86	1.80
	(0.03)	(0.03)	(0.06)	(0.07)
$AFNS_3$	2.01	2.61	1.23	2.28
	(0.04)	(0.01)	(0.22)	(0.02)
Time series models				
HAR	1.41	4.45	0.71	1.72
	(0.16)	(0.00)	(0.48)	(0.09)
Mean-reverting realized GARCH	5.34	10.00	4.23	4.75
	(0.00)	(0.00)	(0.00)	(0.00)

This table shows results from the regression Eq. (22), using pre-averaged realized volatility from high-frequency data as predictor, to test for whether historical volatility contains incremental information about future volatility, relative to the specified models. Reported values are t-statistics for testing $\phi_1^{\tau,h}=0$. Asymptotic p-values in parentheses, based on the Newey-West variance estimator with automatic lag selection of Andrews (1991). The initial estimation period ranges from January 2, 2000, through June 6, 2008, and the out-of-sample period from June 7, 2008, through December 31, 2020.

spanned neither by the parametric term structure models, nor nonparametrically by the yield curve. From the last panel, the latent factor has significant explanatory power at 5% for future forecast errors at two or more maturities for each of the time series models, as well.

5.2.3. The incremental information in historical volatility

In (iii), we investigate whether historical volatility contains information about future volatility not captured by the models considered, using a similar regression as in Section 5.2.1, but with the lagged realized measure based on high-frequency data as predictor (see Andersen and Benzoni (2010) for a related analysis). The specification is Eq. (22), with V_t^{τ} replacing $\hat{F}_t$, and we test for $\phi_1^{\tau,h}=0$.

Results appear in Table 7. According to this test, historical volatility carries significant incremental information at 5% at two or more maturities, relative to each of the term structure models. Similarly, historical volatility contains significant incremental information relative to three or more of the four term structure models at each maturity beside the intermediate $\tau=5$, where the test is only significant for Vasicek. At $\tau=1$, historical volatility carries incremental information relative to all models, including the time series models, possibly due to recursive estimation and regime switching (to ZLB). HAR subsumes the information content in historical volatility at all other maturities, whereas historical volatility carries incremental information relative to realized GARCH across all maturities.

Overall, our specification tests reveal that the term structure of interest rates contains incremental cross-sectional information about future risk at the long end, relative to the time series of historical volatilities. Some of the incremental information in the term structure is not captured by the affine models, and historical volatility carries incremental information relative to these, and to the yield curve itself. The latter phenomenon can be related to the existence of a USV factor, and we find evidence of this in the data.

5.3. Economic value of interest rate risk forecasts

So far, the analysis has focused on statistical measures of predictive ability. We next examine whether the volatility forecasting methods considered generate utility in a portfolio allocation framework, following Bollerslev et al. (2018). To this end, the analysis is translated from the level of yields to returns. We consider an investor purchasing a zero-coupon bond of maturity $\tau+h$ at time t and selling the bond at $t+h$. Let $\tilde{r}_{t+h}$ denote the log return from this trading strategy. The relation between returns and yields is

$$\tilde{r}_{t+h}=-\tau y_{t+h}^{\tau}+(\tau+h)y_t^{\tau+h}, \tag{24}$$

so the conditional variance of the return as of t is given by

$$Var_t(\tilde{r}_{t+h})=\tau^2 Var_t(y_{t+h}^{\tau}), \tag{25}$$

i.e., depending on maturity and the yield volatility forecast we consider. Investor is assumed to have mean-variance preferences and access to a risk-free as well as a risky asset, the latter being the zero-coupon bond with time-varying volatility. Assuming a constant Sharpe ratio, investor's utility depends only on the variance of the risky asset. We consider maturities 7, 13, 61, and 85 months, so that the analysis depends on the forecasts from the previous sections.

Let w_t be the portfolio weight allocated to the risky bond and $1-w_t$ the allocation to the risk-free asset with return r_t^f. The return to the portfolio at time $t+h$ is

$$r_{t+h}=r_t^f+w_t rx_{t+h}, \tag{26}$$

where $rx_{t+h}=\tilde{r}_{t+h}-r_t^f$ is the excess return to the risky asset. Since r_t^f is common across all forecasters, we only consider utility in terms of excess return. Expected utility per unit of wealth is given by

$$\mathcal{U}_{t+h}=w_t\mathbb{E}(rx_{t+h})-\frac{1}{2}\gamma w_t^2 Var_t(rx_{t+h}), \tag{27}$$

where γ is relative risk aversion, and $Var_t(rx_{t+h})=Var_t(\tilde{r}_{t+h})$ from Eq. (25). The optimal weight w_t^* is then

$$w_t^*=\frac{1}{\gamma}\frac{\mathbb{E}_t(rx_{t+h})}{Var_t(rx_{t+h})}, \tag{28}$$

which, given the constant Sharpe ratio, $SR=\mathbb{E}_t(rx_{t+h})/\sqrt{Var_t(rx_{t+h})}$, becomes

$$w_t^*=\frac{1}{\gamma}\frac{SR}{\sqrt{Var_t(rx_{t+h})}}. \tag{29}$$

By Eqs. (26) and (29), the volatility target sought by the investor is $w_t^*\sqrt{Var_t(rx_{t+h})}=SR/\gamma$. If the forecasted volatility $\tau\sqrt{Var_t(y_{t+h}^{\tau})}$ exceeds this target, investor will place only a portion of wealth in the risky asset, $w_t<1$, and save the remainder in the risk-free asset. Conversely, when the forecasted volatility falls short of the target, investor will take a geared position in the risky asset, $w_t>1$, financed by borrowing at the risk-free rate. Thus, investor follows a volatility timing strategy.

To examine the utility gains from the various forecasters, let $\mathbb{E}_t(\cdot)$ denote the conditional expected value from the true model, $\mathbb{E}_t^{\xi}(\cdot)$ the conditional expected value from model ξ, and similarly for conditional variances. Assuming that investor uses model ξ for portfolio selection, the optimal weight becomes $w_{t,\xi}^*=SR/\left(\gamma\tau\sqrt{Var_t^{\xi}(y_{t+h}^{\tau})}\right)$. The expected utility per unit of wealth can then be expressed as

$$\mathcal{U}_{t+h}=\frac{SR^2}{\gamma}\left(\frac{\sqrt{Var_t(y_{t+h}^{\tau})}}{\sqrt{Var_t^{\xi}(y_{t+h}^{\tau})}}-\frac{1}{2}\frac{Var_t(y_{t+h}^{\tau})}{Var_t^{\xi}(y_{t+h}^{\tau})}\right). \tag{30}$$

If a model is able to perfectly predict conditional variance, then investor's expected utility is $SR^2/(2\gamma)$, otherwise it is less. Following Bollerslev et al. (2018), we set the annualized Sharpe ratio to 0.4. The

Table 8
Utility from yield volatility forecasts.

	$\tau = 0.5$	$\tau = 1$	$\tau = 5$	$\tau = 7$
Term structure models				
Vasicek	-3.18	-6.25	0.56	1.15
	(0.97)	(0.89)	(0.04)	(0.61)
CIR	0.50	1.08	0.89	1.18
	(0.75)	(0.41)	(0.02)	(0.53)
$AFNS_0$	0.95	0.58	1.32	1.38
	(0.13)	(1.00)	(0.01)	(0.07)
$AFNS_3$	1.17	0.75	0.95	1.11
	(0.00)	(0.89)	(0.02)	(0.67)
PCA based forecasters				
PCA_3	-1.38	-12.27	1.06	1.04
	(0.91)	(0.89)	(0.02)	(0.79)
PCA_4	-1.54	-11.92	0.51	1.06
	(0.91)	(0.90)	(0.04)	(0.77)
PCA_5	-1.34	-11.93	0.26	1.06
	(0.91)	(0.90)	(0.06)	(0.78)
PCA_6	0.20	-6.65	0.23	1.19
	(0.92)	(0.96)	(0.06)	(0.49)
Risk premium based forecasters				
Forward Spread	0.86	0.58	1.28	1.07
	(0.30)	(1.00)	(0.02)	(0.74)
Yield Spread	0.85	0.57	1.31	1.41
	(0.32)	(1.00)	(0.01)	(0.03)
Common factors based forecasters				
Static	0.02	-4.52	0.69	1.31
	(0.94)	(0.90)	(0.04)	(0.16)
Recursively	-1.51	-12.16	0.34	1.25
	(0.93)	(0.89)	(0.06)	(0.33)
Time series models				
HAR	1.17	0.89	1.32	1.45
	(0.01)	(0.93)	(0.01)	(0.01)
Mean-reverting realized GARCH	0.88	0.62	1.30	1.39
	(0.27)	(1.00)	(0.01)	(0.04)
RW	0.71	1.05	-3.32	1.19
	(-)	(-)	(-)	(-)

This table presents average realized utility from using the specified yield volatility forecasters for portfolio allocation. In parentheses asymptotic p-values for a one-sided Diebold-Mariano test relative to a RW using the Newey-West variance estimator with automatic lag selection of Andrews (1991). A portfolio consisting of the risk-free asset generates utility zero. The initial estimation period ranges from January 2, 2000, through June 6, 2008, and the out-of-sample period from June 7, 2008, through December 31, 2020.

risk aversion parameter, γ, is set to 5, following Sarno et al. (2016) and Gargano et al. (2017) from the bond return prediction literature. This corresponds to an annualized volatility target of 8%.

To evaluate predictors, we simply average realized utility. Table 8 reports the results. They deviate somewhat from those on prediction accuracy, Table 4, presumably because R^2_{OoS} is symmetric, penalizing over- and underprediction equally, whereas the utility criterion penalizes underprediction more heavily. At the $\tau = 5$ year maturity, investor is better off ignoring the RW forecast and investing in the risk-free asset, which generates utility zero. Highest utility at the two longest maturities is achieved using either $AFNS_0$, the yield spread based forecaster, HAR, or mean-reverting realized GARCH, all of which improve significantly over the naive forecast at level 5%, except that $AFNS_0$ gets a p-value of 0.07 at $\tau = 7$. The HAR model generates high utility at the two shortest maturities, as well, along with $AFNS_3$ at $\tau = 0.5$ (both are significant at 1%). The PCA and common factor based forecasters are not quite on par with the term structure models, risk premium based forecasters, and time series models.

Table 9
Yield volatility forecasts using nonlinearities in yield curve factors.

	$\tau = 0.5$	$\tau = 1$	$\tau = 5$	$\tau = 7$
Maturity-specific, initial	-43.25	-38.33	6.07	-13.21
	(0.98)	(0.99)	(0.28)	(0.73)
Maturity-specific, recursive	-28.94	-5.23	4.35	-8.62
	(0.90)	(0.67)	(0.35)	(0.72)
Common factor, initial	-26.31	-17.87	16.15	-17.74
	(0.88)	(0.89)	(0.12)	(0.82)
Common factor, recursive	-23.45	-15.03	17.28	-20.59
	(0.87)	(0.85)	(0.05)	(0.93)

This table displays R^2_{OoS} measures in percent relative to a RW for yield volatility forecasts allowing linear and nonlinear terms as well as interactions in the three leading yield curve factors extracted by PCA in Eq. (8). The optimal combination of terms is selected by BIC. The label "initial" indicates that the selection is based on the initial estimation period, and "recursive" that it is updated every period. In parentheses asymptotic p-values for a one-sided Diebold-Mariano test using the Newey-West variance estimator with automatic lag selection of Andrews (1991). The initial estimation period ranges from January 2, 2000, through June 6, 2008, and the out-of-sample period from June 7, 2008, through December 31, 2020.

Overall, the analysis shows that the information about future reinvestment rate risk contained in both the yield curve (yield spread, affine models) and historical volatility (HAR, mean-reverting realized GARCH) provides economic value to a risk averse investor, both in short and long maturity instruments. There are clear trade-offs, with the affine models, the yield spread, and realized GARCH performing relatively better on utility grounds, and the forward spread in terms of accuracy, along with the PCA based forecaster using three components. The HAR model performs relatively well according to both criterions, forecasting accuracy and utility.

6. Robustness

This section addresses the robustness of the results in Section 5. First, we consider whether interest rate risk forecasts are improved by including nonlinear terms in PCA factors. Next, we examine whether there is a trade-off between explaining yields and forecasting volatility, and we assess the information about future yield volatility in a wide cross section of coupon bond prices. Finally, we explore the possibility of enhancing forecasting performance by combining cross-sectional and intertemporal information.

6.1. Nonlinearities and interest rate risk

Feldhütter et al. (2016) find that including nonlinearities via quadratic and cubic terms as well as interactions in factors increases in-sample explanatory power for realized variance. In our out-of-sample framework, we find in Table 4 that using the three leading factors $\hat{F}_t$ extracted from the yields at time t based on Eq. (8) suffices for the PCA based forecasts, i.e., adding factors beyond three does not increase accuracy. Here, we examine whether including nonlinearities in the first three PCA factors improves forecasts, both for the common factor approach and the maturity-specific forecasts. As in the linear common factor approach, Eq. (9), the selection of terms to include is based on the BIC, either for the initial estimation period, or updated recursively.

Results appear in Table 9. Recursively updating the selection of the nonlinear terms generates more accurate forecasts than selecting based on the initial estimation period, only, except at the longest maturity for the common factor approach, and at $\tau = 5$ for the maturity-specific forecasts.[14] The common factor based forecasts are only more precise than the maturity-specific at two maturities, in contrast to all maturities in the linear case, Table 4. Comparing Tables 4 and 9, including nonlinear terms improves the maturity specific forecasts at all maturities, except $\tau = 5$, and the common factor based forecasts at the two shorter horizons.

Overall, the best forecast across both the maturity-specific and common factor approaches including nonlinearities improves on the linear forecasts for the two shortest maturities, not for the two longest. In short, allowing for nonlinearities does not generate an overall improvement in yield volatility forecasting.

6.2. Trade-off between explaining yields and forecasting volatility

The finding in Table 4 that using the first three PCA factors $\hat{F}_t$ from Eq. (8) generates more accurate yield volatility forecasts than using the first four, five, or six factors raises the additional question of whether using exactly these three is optimal, or whether some of the remaining factors beyond the first three are more informative about future yield volatility. Further, if volatility forecasts are indeed improved by using some other combination of factors, the question arises whether the optimal combination is common across maturities.

To address these issues, we consider a regression of the type in Eq. (1), with Z_t representing a combination of three of the first six PCA factors, i.e., not necessarily the first three. We consider all 20 possible combinations.

Table 10 presents the R^2_{OoS} results. Strongest overall forecasting performance is obtained at the intermediate maturity $\tau = 5$ years. This echoes the result from Table 4, where forecasting based on the first three factors, $\hat{F}_t = \text{PCA}_{123}$, say, improves significantly over the RW at level 5% for $\tau = 5$. From Table 10, nine of the 20 factor combinations improve significantly over the RW at $\tau = 5$, and of these, the first three factors (bottom row of table) generate the lowest R^2_{OoS} statistic, along with PCA_{126}. The best combination (highest statistic) is PCA_{245}, i.e., combining the second, fourth, and fifth PCA factors from Eq. (8) for purposes of yield volatility forecasting. The second factor, slope, enters all of the nine significant combinations, and so proves important for volatility forecasting at $\tau = 5$.[15]

Although significant, the R^2_{OoS} statistic generated by the standard PCA_{123} combination only ranks 13 in magnitude out of the 20 cases at $\tau = 5$, i.e., in the lower half. It ranks last at the short end, $\tau = 0.5$, second to last at $\tau = 1$, and again 13 at the long end, $\tau = 7$. At $\tau = 1$, the best combination is PCA_{456}, i.e., nonoverlapping with the three standard factors. The sixth factor, PCA_6, say, enters the six best combinations, and slope, PCA_2, only two of these. At $\tau = 0.5$, the best combination is PCA_{246}. The sixth factor, PCA_6, enters the five best combinations, and PCA_2 and PCA_5 each three of these. At $\tau = 7$, PCA_{236} is the best combination, PCA_6 enters four of the five best combinations, and PCA_2 and PCA_3 each three of these.

Summing up, the evidence is clearly that using the three leading principal components is not optimal for purposes of forecasting yield volatilities. The combination improves over the RW at the intermediate maturity $\tau = 5$ years, but it is in the lower half of factor combinations in terms of forecasting accuracy at each maturity. Further, the optimal combination of factors for volatility forecasting is not common across maturities. The slope factor is important at $\tau = 5$, and enters the best combination at all maturities except $\tau = 1$, where it enters the second best. Higher-order factors, the fourth through sixth principal components, enter the best combinations for volatility forecasting purposes at all maturities. The sixth factor is particularly important. It enters the best combination at all maturities except $\tau = 5$, and the five best combinations at each of the two shortest maturities.

The deterioration in performance by adding factors beyond the leading three in Table 4 indicates the importance of parsimony. Although higher-order factors are informative about volatility, this is only re-

Table 10
Trade-off between explaining yields and forecasting volatility.

	$\tau = 0.5$	$\tau = 1$	$\tau = 5$	$\tau = 7$
PCA_{456}	-19.01	-13.69	4.22	-41.89
	(0.86)	(0.85)	(0.38)	(0.97)
PCA_{356}	-22.31	-17.47	17.48	-8.88
	(0.88)	(0.88)	(0.07)	(0.69)
PCA_{346}	-26.34	-20.44	15.60	-6.01
	(0.91)	(0.93)	(0.10)	(0.66)
PCA_{345}	-26.03	-18.65	17.39	-8.17
	(0.88)	(0.87)	(0.08)	(0.72)
PCA_{256}	-19.85	-14.31	18.22	-9.91
	(0.86)	(0.85)	(0.04)	(0.75)
PCA_{246}	-18.78	-15.91	15.70	-4.57
	(0.86)	(0.88)	(0.07)	(0.68)
PCA_{245}	-23.85	-18.36	21.53	-18.82
	(0.85)	(0.90)	(0.01)	(0.94)
PCA_{236}	-23.30	-20.03	16.18	-0.05
	(0.89)	(0.91)	(0.05)	(0.50)
PCA_{235}	-27.27	-20.22	20.72	-12.4
	(0.88)	(0.88)	(0.01)	(0.87)
PCA_{234}	-27.48	-20.92	18.82	-12.21
	(0.90)	(0.90)	(0.02)	(0.86)
PCA_{156}	-24.31	-17.03	6.14	-36.16
	(0.90)	(0.88)	(0.35)	(0.92)
PCA_{146}	-24.20	-18.35	6.41	-26.83
	(0.91)	(0.91)	(0.32)	(0.93)
PCA_{145}	-27.88	-20.19	6.59	-34.79
	(0.89)	(0.91)	(0.34)	(0.97)
PCA_{136}	-29.18	-24.92	17.29	-0.53
	(0.93)	(0.95)	(0.08)	(0.51)
PCA_{135}	-31.09	-23.02	16.80	-5.82
	(0.91)	(0.90)	(0.11)	(0.64)
PCA_{134}	-31.68	-23.86	17.38	-2.42
	(0.92)	(0.91)	(0.08)	(0.58)
PCA_{126}	-24.15	-19.05	15.91	-3.71
	(0.90)	(0.90)	(0.04)	(0.61)
PCA_{125}	-29.97	-21.76	18.57	-21.63
	(0.89)	(0.92)	(0.05)	(0.91)
PCA_{124}	-28.09	-21.89	19.21	-15.59
	(0.89)	(0.92)	(0.02)	(0.89)
PCA_{123}	-32.11	-24.35	15.92	-15.84
	(0.92)	(0.91)	(0.04)	(0.91)

This table displays R^2_{OoS} measures in percent relative to a RW for all combinations of three of the first six PCA yield factors from Eq. (8). In parentheses asymptotic p-values for a one-sided Diebold-Mariano test using the Newey-West variance estimator with automatic lag selection of Andrews (1991). The initial estimation period ranges from January 2, 2000 to June 6, 2008, and the out-of-sample period from June 7, 2008, through December 31, 2020.

[14] In contrast, initial selection dominates updating for the linear common factor based forecasts in Table 4.

[15] The final combination involving slope, PCA_{246}, gets p-value 0.07.

vealed by including these in parsimonious combinations, in our case using three factors at a time. The results indicate a trade-off between explaining yield levels and forecasting interest rate risk. The three leading principal components from the yield panel are known to capture current yield curves well, but the fourth and higher components are important for forecasting yield volatilities.

6.3. Interest rate risk forecasts based on a cross section of coupon bond prices

The estimation of our models relies on panel data on yields constructed from high-frequency futures prices, with five price observations in the cross section dimension, for the time series methods, and on the daily panel with $N = 8$ yields in the cross section for the curve-based methods. Here, we investigate the possibility that the full cross section of observed coupon bond prices at any point in time carries more information about future interest rate risk. We consider daily CRSP data on quoted prices of bonds with maturities between 3 months and 10 years, adjusted for accrued interest, and excluding callable issues. The price of a bond promising M semi-annual payments of $C/2$ with maturities $\tau = (\tau_1, \dots, \tau_M)$ is represented as

$$P_t(\tau, C) = \frac{C}{2}\frac{\tau_1}{1/2}B_t^{\tau_1} + \sum_{j=2}^{M}\frac{C}{2}B_t^{\tau_j} + 100B_t^{\tau_M}, \tag{31}$$

with $B_t^{\tau_j} = \exp(-\tau_j y_t^{\tau_j})$ the price of a zero-coupon bond of maturity τ_j, and $\tau_1/(1/2)$ the fraction of the first coupon receivable, reflecting the share of the half-year interval before the nearest coupon over which the coupon bond is held. Bond prices $P_t^{i,obs}$ are assumed to be observed with measurement error,

$$P_t^{i,obs} = P_t(\tau^i, C^i) + \varepsilon_t^i, \tag{32}$$

with $P_t(\cdot)$ the model-implied price from Eq. (31), (τ^i, C^i) contractual terms for bond i, and the measurement error ε_t^i Gaussian with variance σ_t^2, independently across observations $i = 1, \dots, N_t$ available at t. Write P_t^{obs} for the N_t-vector of observed coupon bond prices, Θ_t for the $\mathbb{Q}$-parameters of the pricing model, augmented with the short rate r_t, considered latent and backed out from the cross section, and $P(\Theta_t)$ for the N_t-vector of model-implied prices. Estimation is by cross-sectional nonlinear regression or QML,

$$\hat{\Theta}_t = \arg\max_{\Theta_t}\left(-\frac{N_t}{2}\log\sigma_t^2 - \frac{1}{2\sigma_t^2}(P_t^{obs} - P(\Theta_t))'(P_t^{obs} - P(\Theta_t))\right). \tag{33}$$

The cross-sectional estimate $\hat{\Theta}_t$ does not depend on σ_t^2. Fig. A.2 in the Appendix shows the evolution over time in N_t, ranging from a low of 85 in the early part of the sample to a high of 280 towards the end.

For purposes of yield volatility forecasting, $\mathbb{Q}$-parameters do not suffice, as $V_{t,h}^{\tau}$ in Eq. (7) depends on $\mathbb{P}$-parameters through the coefficients $b_0^{\tau,h}$ and $b_1^{\tau,h}$. For concreteness, we focus on the CIR model, i.e., Eq. (33) is the estimator considered by Brown and Dybvig (1986), who examined the consistency between the cross-sectional estimate of the short rate volatility parameter σ in the square root process and a time series estimate of this, based on the sample variance of changes in estimated short rates r_t across calendar time. This comparison was relevant because σ is common across $\mathbb{Q}$ and $\mathbb{P}$ in the continuous-time model (see Appendix C.2). Instead, since we consider volatility forecasting at longer maturities, we estimate the market price of risk λ (cf. Table 1) from a time series of short rates calibrated period by period in Eq. (33). Specifically, we consider an Euler discretization of the CIR process,

$$r_{t+1} - r_t = \kappa(\theta - r_t) + \sigma\sqrt{r_t}\varepsilon_{t+1}, \tag{34}$$

following Chan et al. (1992), who focused on the $\mathbb{P}$-parameter estimates resulting from time series analysis of Eq. (34). Given the estimated $\mathbb{Q}$-parameters from Eq. (33), we instead express Eq. (34) as a function of λ. Since r_{t+1} conditional on r_t is Gaussian under the discretization, we estimate λ by QML. For forecasts at t, we use $\mathbb{Q}$-parameters from Eq. (33) at t, along with λ estimated from a short time series $t-10, \dots, t$ in Eq. (34). An alternative would be to estimate all three $\mathbb{P}$-parameters in a longer time series, and construct final estimates of all four parameters (including λ) by optimal weighting of estimates from the sequential cross-sectional and time series regressions, following Andreasen and Christensen (2015). Rather than pursuing this approach, we focus on the short time series of fitted r_t for setting λ, to retain the cross-sectional nature of the approach.

Table 11
Yield volatility forecasts from cross section of coupon bond prices.

	$\tau = 0.5$	$\tau = 1$	$\tau = 5$	$\tau = 7$
$\text{CIR}_{\mathbb{Q}}$	-26.64	-16.21	10.58	-33.03
	(0.98)	(0.90)	(0.03)	(1.00)
$\text{CIR}_{\mathbb{P}}$	-30.20	-20.13	10.60	-33.92
	(0.99)	(0.94)	(0.03)	(1.00)

This table displays R^2_{OoS} measures in percent relative to a RW for the CIR model estimated on a cross section of coupon bonds. $\text{CIR}_{\mathbb{Q}}$ indicates that forecasting is based on cross-sectionally estimated $\mathbb{Q}$-parameters and short rates, and $\text{CIR}_{\mathbb{P}}$ that the market price of risk is estimated based on the $\mathbb{Q}$-parameters, the ten most recent fitted short rates, and Euler discretization. In parentheses asymptotic p-values for a one-sided Diebold-Mariano test using the Newey-West variance estimator with automatic lag selection of Andrews (1991). The initial estimation period ranges from January 2, 2000 to June 6, 2008, and the out-of-sample period from June 7, 2008, through December 31, 2020.

Table 11 shows the R^2_{OoS} results. For comparison, statistics are shown both for forecasts based purely on cross-sectionally estimated $\mathbb{Q}$-parameters and short rates, and for the combined approach described. The former are highest across all maturities, presumably due to parsimony. They are slightly higher than the corresponding statistics for the CIR model based on the daily yield panel in Table 4 at the shorter maturities, and lower at the longer. The results suggest that the wide cross section is relatively useful for volatility forecasting at the short end of the curve, possibly because all instruments include nearby coupons. Further, at $\tau = 5$, although R^2_{OoS} is slightly higher in the yield panel than in the cross section of coupon bonds, the improvement over the RW is significant at 5% in the latter case, both for the $\mathbb{Q}$-based and the combined approach, not in the former (p-value of 19%). This extends the relative usefulness of the forecasts from the wide cross section to all maturities but the longest, where it performs poorly, both in absolute terms, and compared to the yield panel. In sum, the approaches based on the yield panel and the cross section of coupon bonds exhibit broadly similar volatility forecasting performance, with a small advantage to the former at the long end, and to the latter at shorter maturities.

Overall, the robustness analysis confirms that the daily yield panel captures the cross-sectional information adequately. Further, more of the future volatility information in the curve not captured by standard term structure models appears to be contained in higher-order factors than in nonlinearities.

6.4. Time series extensions

So far, the analysis has focused on simple, linear specifications for the time series models. However, the literature on modeling stock return volatility is rich on extensions, e.g., to the basic HAR model in Eq. (10). We now examine some extensions within the HAR framework. First, we allow for a nonlinear relation between past and future volatility by specifying Eq. (10) in logarithmic terms. Each realized measure

Table 12
Time series extensions.

	$\tau = 0.5$	$\tau = 1$	$\tau = 5$	$\tau = 7$
	Panel A: HAR extensions			
Log-HAR	4.05	6.69	17.08	14.38
	(0.39)	(0.30)	(0.02)	(0.11)
Lev-HAR	-5.40	-0.64	22.01	14.35
	(0.61)	(0.52)	(0.00)	(0.12)
	Panel B: Interest rate HAR extensions			
PC1	-0.15	6.88	21.50	12.80
	(0.50)	(0.31)	(0.01)	(0.09)
PC2	0.29	7.08	25.31	11.58
	(0.49)	(0.31)	(0.00)	(0.07)
PC3	-0.86	5.33	24.66	18.04
	(0.52)	(0.35)	(0.00)	(0.02)
PC1 interaction	0.12	9.35	21.93	12.84
	(0.50)	(0.23)	(0.00)	(0.07)
PC2 interaction	-4.09	10.03	26.54	11.96
	(0.62)	(0.22)	(0.00)	(0.05)
PC3 interaction	-10.13	0.29	23.86	18.91
	(0.69)	(0.49)	(0.00)	(0.01)

This table displays R^2_{OoS} measures in percent relative to a RW for the Log-HAR model, the Lev-HAR model, and the HAR model with time-varying parameters, with either intercept or both intercept and slopes depending in an affine manner on one of the three leading principal components from the daily yield panel. In parentheses asymptotic p-values for a one-sided Diebold-Mariano test using the Newey-West variance estimator with automatic lag selection of Andrews (1991). The initial estimation period ranges from January 2, 2000 to June 6, 2008, and the out-of-sample period from June 7, 2008, through December 31, 2020.

is replaced by its logarithm, for estimation purposes, and a Jensen's inequality correction applied when constructing level forecasts (exponentiating the relevant fitted value plus one half times the residual variance). We label this the Log-HAR model. Second, stocks are affected by the so-called leverage effect. We examine whether this is also the case for interest rates and, in particular, whether it matters for yield volatility forecasting, by considering the leverage HAR (or Lev-HAR) model of Corsi and Renò (2012). The model allows for a leverage effect by extending Eq. (10) with the sum of past negative returns. It is given as

$$V^{\tau}_{t+1\to t+h} = \beta^{\tau,h}_0 + \beta^{\tau,h}_D V^{\tau}_t + \beta^{\tau,h}_W V^{\tau}_{t-4\to t} + \beta^{\tau,h}_M V^{\tau}_{t-21\to t} + \gamma^{\tau,h}_D |r^-_{t,\tau,1}| + \gamma^{\tau,h}_W |r^-_{t,\tau,5}| + \gamma^{\tau,h}_M |r^-_{t,\tau,22}| + u^{\tau,h}_{t+h}, \tag{35}$$

where $r^-_{t,\tau,h}$ aggregates any negative changes in the maturity τ yield over the past h days.

Panel A in Table 12 presents the R^2_{OoS} results. Comparing with the standard HAR model in Table 4, each extension generates higher R^2_{OoS} at two maturities ($\tau = 0.5$ and 7 for Log-HAR, $\tau = 5$ and 7 for Lev-HAR), and lower than HAR at the other two. All three models improve significantly over the RW at maturity $\tau = 5$ and level 5%, and get p-values around 11% for the longest maturity. Thus, the evidence on the extensions is mixed, and the standard HAR model does not appear to be a bad choice. Regarding the leverage effect, a potential explanation is that compared to returns, yields tend to have a truncation around zero, as in shadow rate models (Black, 1995; Christensen and Rudebusch, 2015; Andreasen and Meldrum, 2019). The truncation lowers yield volatility for yields near zero by construction. Thus, we do not necessarily expect a leverage effect in interest rates.

Overall, our results show that either information in the yield curve or in the volatility time series can be used to improve yield volatility forecasting. The question arises whether forecasting performance can be further enhanced by combining the two sources of information. We address this issue in the HAR framework via a time-varying parameter extension. In particular, we allow for the HAR coefficients to depend on information from the yield curve. This notion seems natural, as interest rates are known to carry important economic signals. Thus, the yield spread is widely considered to be a recession indicator, and the interest rate level measures the distance to the ZLB. Furthermore, we have shown that interest rate factors can forecast the volatility forecasting errors from the HAR model (Section 5.2.1). This suggests that interest rate information can be incorporated in the HAR framework in a productive manner. For a simple specification, we consider in turn whether each of the three first PCA factors from the yield curve drives variation in the intercept in the HAR model, or in both intercept and slopes, for an interactive effect between yield curve and volatility information. Here, the first PCA factor, level, should capture distance to the ZLB, and the next two, slope and curvature, the effect of the spread. Thus, in the generalized models, when Eq. (10) is estimated based on information through t', the extended parameter specification

$$\beta^{\tau,h}_{i,t} = \alpha^{\tau,h}_i + \gamma^{\tau,h}_i \hat{F}_{t,j} \tag{36}$$

is used, with $\hat{F}_{t,j}$, $j = 1, 2, 3$ the fitted yield curve factors at $t \le t'$ from estimation of Eq. (8) with $k = 3$, using data through t'. In the time-varying intercept extensions, Eq. (36) is only used for $i = 0$. In the interactive extensions, it is used for $i = 0, D, W, M$.

Other approaches to time-varying parameters in the HAR framework have been proposed in the literature, e.g., the HARQ model of Bollerslev et al. (2016), and the model of Buccheri and Corsi (2021). However, these models rely on the basic realized volatility framework and, hence, are not naturally transferred to our pre-averaging estimator setting. Therefore, we do not pursue these alternative approaches further.

Panel B in Table 12 shows the R^2_{OoS} results from using Eq. (36) with Eq. (10). At the longest maturities, extending the HAR model with information from the yield curve enhances volatility forecasting performance, relative to the standard model from Table 4. Allowing for interactions using level or slope, performance is improved at $\tau = 1$, too. For given maturity, the best volatility forecasts considered are those using the original HAR at the short end, the interactive HAR extension with slope at the two intermediate maturities, and that with curvature at the long end of the curve. The results suggest that recession risk is important at longer maturities. All extensions retain significance at level 1% of the improvement over the RW at $\tau = 5$ and, in contrast to the standard HAR, deliver significant improvements at the longest maturity, too, at 10%. The time-varying intercept model using curvature and the interactive model using slope provide significant improvements at 5%, and the interactive model using curvature at 1%. Obviously, further research could investigate combinations using multiple factors, but the results confirm that cross-sectional yield curve information and historical time series information on volatility can be fruitfully combined for volatility forecasting purposes.

7. Concluding remarks

Our results show that the assessment of future interest rate risk is a complicated affair. A strong benchmark can be formed by basing forecasts on historical volatility, with a focus on recent periods, as in the formal time-varying volatility models. A good alternative at intermediate maturities can be based on the risk premium, in particular, the forward spread, or a common factor approach. For agents with a primary interest in the long end of the curve, e.g., for immunization of assets and liabilities, valuation of capital assets, or market timing, incremental information on future yield volatility relative to that in the

time series is available by looking across maturities along the curve, and more so than implied by formal term structure models. Some of the curve-based information about future volatility is distinct from that explaining the levels of yields, and is contained in higher-order factors rather than nonlinearities. Furthermore, our results point to the existence of a latent stochastic volatility factor, unspanned by yields.

Based on a simple portfolio exercise, the information about future reinvestment rate risk in either the yield curve or the volatility history is of economic value to a risk averse investor, both in short and long term instruments. A simple yield spread, affine models, and time series methods are all useful, on utility grounds, relative to a random walk. Finally, we demonstrate that the cross-sectional yield curve information and the historical time series information on volatility can be fruitfully combined to enhance volatility forecasting performance.

We have emphasized from the beginning that when leveraging volatility information drawn from the yield curve, investor should not ignore historical volatility in the information set. That the resulting picture is diverse is perhaps not surprising. Investor is, after all, gazing into the crystal ball. Clearly, mimicking this situation requires a recursive, out-of-sample approach, to avoid admitting the artificial investor the benefit of hindsight.

CRediT authorship contribution statement

All authors have contributed equally to all parts of the article. No other authors have been involved.

Declaration of competing interest

None.

Data availability

Data will be made available on request.

Appendix A. From futures prices to yield curves

A.1. From futures prices to coupon bond prices

The seller (the short) has two options included in the Treasury futures. The first regards which bond to deliver, and the second the delivery date within the delivery month. For the first option, the underlying assets of the Treasury futures (see Section 3) are hypothetical bonds with a notional yield of 6% throughout our sample period. The seller must deliver an actual bond, selected from a delivery basket constructed for each futures contract according to CME requirements, with a conversion factor for each bond. The conversion factor converts the bond price into "... the approximately decimal price at which \$1 par of the security would trade as if it had a 6% yield-to-maturity."[16] The formula is

$$f = a\left(\frac{coupon}{2} + c + d\right) - b \tag{A.1}$$

$$a = \left(\frac{1}{1.03}\right)^{\frac{v}{6}}$$

$$b = \frac{coupon}{2}\frac{6-v}{6}$$

$$c = \begin{cases} \left(\frac{1}{1.03}\right)^{2n} & \text{if } z < 7 \\ \left(\frac{1}{1.03}\right)^{2n+1} & \text{otherwise} \end{cases}$$

$$d = \frac{coupon}{0.06}(1-c)$$

$$v = \begin{cases} z & \text{if } z < 7 \\ 3 & \text{if } z \geq 7 \text{ for 10-year note, long term bond} \\ z-6 & \text{if } z \geq 7 \text{ for 5-year note} \end{cases}$$

with *coupon* the annualized coupon, n the number of (whole) years from the first day of the delivery month to maturity, and z the number of months between n and the maturity date, rounded to nearest quarter for the 10-year note and the long term Treasury bond, and nearest month for the 5-year note. The delivery bond (or note) must have a maturity between 4.17 and 5.25 years for the 5-year futures, between 6.5 and 10 years for the 10-year futures, and between 15 and 25 years for the long term Treasury bond futures.

Given the conversion factor f_t from Eq. (A.1), the invoice amount is

$$I_t = f_t \cdot F_t + a_t, \tag{A.2}$$

with F_t the futures price, and a_t accrued interest. On delivery, the seller pays the basis

$$\pi_t = S_t - I_t, \tag{A.3}$$

with S_t the cash bond price (including accrued interest), and therefore selects the cheapest-to-deliver (CTD) bond fulfilling the futures contract specifications and minimizing Eq. (A.3) (maximizing the implied repo rate).

The second option regards the delivery date. For the purpose of backing out the futures-implied bond price, we assume that both the CTD bond and delivery date are known. We set the delivery date to the first working day of the delivery month, and use the CRSP data on Treasuries to construct the delivery basket. The price of the futures is then given as

$$I_t = (S_t - C_t)e^{r_t T}, \tag{A.4}$$

with S_t the spot price of the CTD bond (including accrued interest), C_t the present value of coupon payments before delivery, T the time to delivery, and r_t the risk-free rate. For the latter, we use the 3-month yield from the daily panel, $r_t = y_t^{1/4}$ (Section 3.5). The 3-month rate should be more market-based than, say, a 1-month rate, and a good short rate proxy, cf. Chapman et al. (1999).

In brief, the recipe for constructing the futures-implied bond price S_t is as follows:

(i) Construct the delivery basket from the CRSP data on the first trading day of the delivery month.
(ii) Find the cheapest-to-deliver bond minimizing the basis π_t from Eq. (A.3) within the delivery basket from (i).
(iii) Use Eq. (A.2) to calculate the invoice amount I_t for the CTD bond from (ii).
(iv) Substitute the invoice amount from (iii) for I_t in Eq. (A.4), and back out the futures-implied bond price by isolating S_t.

A.2. From coupon bond prices to yield curves

A variety of methods is available for extracting the yield curve from the cross section of coupon bond prices. In the literature on yield curves at high frequency, some version of cubic spline is usually employed, e.g., Andersen and Benzoni (2010) and Cieslak and Povala (2016), due to the small number of observations in the cross section, and the need for flexibility to fit the curve reasonably. As a consequence, interpolation between points of principal payments is difficult. Following Cieslak and Povala (2016), we use the method of Fisher et al. (1995). This assumes that the discount function $B_t^\tau = \exp(-\tau y_t^\tau)$ is determined by a cubic spline $h(\cdot, \Psi)$, i.e., $\tau y_t^\tau = h(\tau, \Psi)$, with Ψ a set of parameters (suppressing dependence on t). The yield curve is split along the maturity axis using K knot points, $0 < \tau_1 < \ldots < \tau_K$, with τ_K the maximum maturity of the bonds considered. The cubic spline is restricted such that $h(\cdot, \Psi)$ and its two first derivatives are continuous. This implies that one

[16] https://www.cmegroup.com/trading/interest-rates/calculating-us-treasury-futures-conversion-factors.html.

Table A.1
Correlation between futures-implied and Gürkaynak et al. (2007) yields.

$\tau = 0.5$	$\tau = 1$	$\tau = 5$	$\tau = 7$
99.99%	99.99%	99.62%	99.37%

This table shows the sample correlation between end-of-day futures-implied yields and yields constructed using Gürkaynak et al. (2007) parameter estimates. The sample spans the period from January 2, 2000, through December 31, 2020.

parameter is added for each additional knot point. Fisher et al. (1995) and Dai et al. (2007) set K to approximately one third of the number of bonds included, while Andersen and Benzoni (2010) and Cieslak and Povala (2016) set K equal to the number of bonds. We follow the high-frequency literature and set the number of knot points equal to the number of bonds on which we calibrate the yield curve, i.e., $K = 5$. We let $\{s_k\}_{k=1}^{K}$ denote the set of knot points.

To calibrate $h(\tau, \Psi)$, we follow Fisher et al. (1995) and use the simple parametrization of the cubic B-spline basis. Any cubic spline can be constructed as a linear combination of B-splines, $h(\tau, \Psi) = \phi(\tau)\Psi$, where Ψ is a $\kappa \times 1$ vector of coefficients, and $\phi(\cdot)$ is a cubic B-spline basis, i.e., a vector of $\kappa = K + 2$ cubic B-splines satisfying

$$\phi_k^r(\tau) = \phi_k^{r-1}(\tau)\frac{\tau - d_k}{d_{k+r-1} - d_k} - \phi_{k+1}^{r-1}(\tau)\frac{d_{k+r} - \tau}{d_{k+r} - d_{k+1}}.$$

Here, $r = 4$ for a cubic spline, and d_k is an augmented set of knot points, $d_1 = d_2 = d_3 = s_1$, $d_{k+3} = s_k$ for $1 \le k \le \kappa$, and $d_{\kappa+4} = d_{\kappa+5} = d_{\kappa+6} = s_\kappa$.

The price of a coupon bond is written as

$$P(\Psi) = \sum_{j=1}^{J} C_j \exp(-h(t_j, \Psi)),$$

with J the number of payments (coupons and principal) over the remaining lifetime of the bond, and C_j the jth payment, due t_j periods hence. The yield curve is estimated using penalized nonlinear least squares (PNLS),

$$\hat{\Psi} = \arg\min_{\Psi} \sum_{i=1}^{K} (P_i - \hat{P}_i(\Psi))^2 + \lambda \int_0^{\tau_K} h''(s)^2 ds,$$

where the penalty term with coefficient λ induces smoothness of the calibrated yield curve. We find that this procedure results in a better fit to bond prices than the Nelson and Siegel (1987) approach used in Faust et al. (2007).

Upon estimation, four points are read off the resulting high-frequency (1 minute) yield curves, at $\tau = 0.5$, 1, 5, and 7 years.

Appendix B. Liquidity of Treasury futures

Table A.2 shows the percentage of high-frequency intervals containing a Treasury futures price observation, and hence affording unique yield identification, at the 1, 5, and 10 minute sampling frequencies, by maturity. All three maturities, 5 and 10 years, and long term, are included in the table, because all three Treasury futures are used in curve fitting, although only implied yields at maturities up to $\tau = 7$ are used in the subsequent analysis (see Section 3.3 and Appendix A). From the table, around 86% to 95% of the 1 minute intervals contain an observation. This increases to around 97% and 99% for 5 and 10 minute intervals, respectively. The lower panel of the table shows the corresponding percentages when discarding trading days with no trading activity for more than an hour. All numbers increase by around 0.5-1.0%. This suggests that the discarded days do not contain many uniquely identified observations and, hence, it makes sense to discard them. Fig. A.1 shows the evolution over time in the daily percentage

Table A.2
Liquidity of Treasury futures.

	1-min	5-min	10-min
	Full sample		
5-year note	86.45%	97.10%	98.76%
10-year note	95.10%	99.14%	99.30%
Long term bond	95.41%	99.27%	99.37%
	Trading days with no activity for more than an hour discarded		
5-year note	87.10%	97.64%	99.27%
10-year note	95.69%	99.67%	99.85%
Long term bond	95.97%	99.77%	99.86%

This table shows the fraction of high-frequency intervals containing a Treasury futures price observation, by underlying maturity and sampling frequency. The full sample is included in the upper panel. In the lower panel, days with no trading activity for more than one hour are discarded. The sample spans the period from January 2, 2000, through December 31, 2020.

of intervals containing a futures price observation. There is a jump between 2003 and 2004, due to the introduction of the electronic trading pit in 2004, and clearly fewer intervals containing futures price observations at the 1 minute frequency than at 5 and 10 minutes, at least until 2008. This suggests that the high-frequency Treasury futures data are likely afflicted with market microstructure noise.

Appendix C. Models

C.1. The Vasicek model

The model of Vasicek (1977) is an $A_0(1)$ model in which the dynamics of the short rate $r_t = y_t^0$ are given by

$$dr_t = \kappa(\theta - r_t)dt + \sigma dW_t.$$

We assume that the market price of risk is completely affine, i.e., $\lambda_t = \lambda$. Thus,

$$\tilde{\theta} = \theta - \frac{\sigma\lambda}{\kappa}, \quad \tilde{\kappa} = \kappa.$$

The solution to the Riccati equations (6) is given by

$$B(\tau) = \frac{1}{\tilde{\kappa}}(1 - e^{-\tilde{\kappa}\tau}),$$

$$A(\tau) = \left(\tilde{\theta} - \frac{\sigma^2}{2\tilde{\kappa}^2}\right)(\tau - B(\tau)) + \frac{\sigma^2}{4\tilde{\kappa}}B(\tau)^2.$$

The conditional variance of the future short rate is

$$Var_t(r_{t+h}) = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa h}).$$

C.2. The Cox-Ingersoll-Ross model

The model of Cox et al. (1985) is an $A_1(1)$ model with

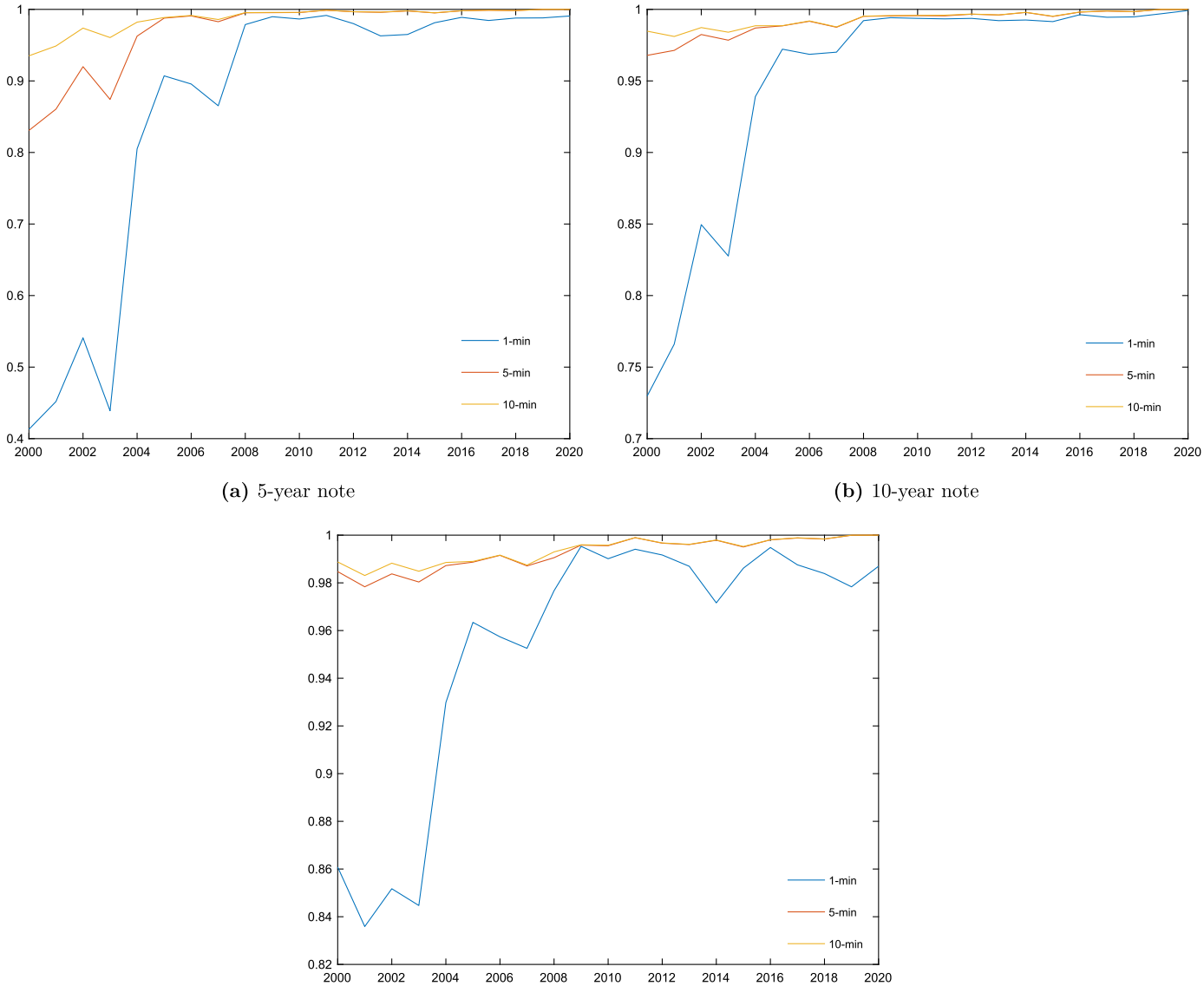
$$dr_t = \kappa(\theta - r_t)dt + \sigma\sqrt{r_t}dW_t.$$

We adopt the completely affine market price of risk specification, $\lambda_t = \lambda\sqrt{r_t}/\sigma$. Thus,

$$\tilde{\kappa} = \kappa + \lambda, \quad \tilde{\theta} = \frac{\kappa\theta}{\kappa + \lambda}.$$

The solution to the Riccati equations (6) is given by

$$B(\tau) = \frac{2(e^{\gamma\tau} - 1)}{(\gamma + \tilde{\kappa})(e^{\gamma\tau} - 1) + 2\gamma},$$



(a) 5-year note

(b) 10-year note

(c) Long term bond

Fig. A.1. Liquidity of Treasury futures over time.
This figure shows the evolution in the daily fraction of high-frequency intervals containing a Treasury futures price observation, by underlying maturity and sampling frequency. The sample spans the period from January 2, 2000, through December 31, 2020.

$$A(\tau) = -\frac{2\tilde{\kappa}\tilde{\theta}}{\sigma^2}\left[\log(2\gamma) + \frac{1}{2}(\tilde{\kappa}+\gamma) - \log\left((\gamma+\tilde{\kappa})(e^{\gamma\tau}-1)+2\gamma\right)\right],$$

with γ given by

$$\gamma = \sqrt{\tilde{\kappa}^2 + 2\sigma^2}.$$

The conditional variance of the short rate is

$$Var_t(r_{t+h}) = \frac{\sigma^2 r_t}{\kappa}(e^{-\kappa h} - e^{-2\kappa h}) + \frac{\sigma^2\theta}{2\kappa}(1-e^{-\kappa h}).$$

C.3. The arbitrage-free Nelson-Siegel model with deterministic volatility

The AFNS_0 model of Christensen et al. (2011) is an $A_0(3)$ model. The real-world dynamics of the state variables are given by

$$dX_t = \kappa(\theta - X_t)dt + \Sigma dW_t,$$

where X_t and θ are 3×1 vectors, and κ and Σ are 3×3 matrices. The model is constructed to make the shape of the yield curve resemble the Nelson and Siegel (1987) parametrization by assuming that the market price of risk is essentially affine and restricting

$$\tilde{\kappa} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -\lambda & 0 \\ 0 & \lambda & \lambda \end{pmatrix},$$

$$\tilde{\theta} = 0_{3\times 1}.$$

The yield curve is then

$$y_t^\tau = X_t^1 + \frac{1-e^{-\lambda\tau}}{\lambda\tau}X_t^2 + \left(\frac{1-e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau}\right)X_t^3 - \frac{A(\tau)}{\tau}. \tag{C.5}$$

This corresponds to the Nelson and Siegel (1987) shape, except for the term $\frac{A(\tau)}{\tau}$, for which the closed form is given in Christensen et al. (2011).

Table A.3
Out-of-sample R^2 for two months ahead yield volatility forecasting.

	$\tau = 0.5$	$\tau = 1$	$\tau = 5$	$\tau = 7$
Term structure models				
Vasicek	17.09	19.25	-35.37	-131.46
	(0.25)	(0.21)	(0.93)	(1.00)
CIR	-11.77	-10.38	-23.05	-143.81
	(0.66)	(0.65)	(0.84)	(0.95)
$AFNS_0$	-13.57	-33.48	6.01	-67.70
	(0.71)	(0.94)	(0.38)	(0.98)
$AFNS_3$	2.28	-4.91	-26.45	-135.69
	(0.45)	(0.56)	(0.86)	(1.00)
PCA based forecasters				
PCA_3	-2.83	-12.78	-12.39	-95.14
	(0.57)	(0.69)	(0.75)	(1.00)
PCA_4	-5.65	-14.08	-16.98	-99.65
	(0.64)	(0.72)	(0.82)	(1.00)
PCA_5	-7.76	-14.21	-26.33	-128.23
	(0.69)	(0.72)	(0.85)	(1.00)
PCA_6	-19.98	-25.55	-28.44	-143.78
	(0.86)	(0.88)	(0.87)	(1.00)
Risk premium based forecasters				
Forward spreads	7.32	-1.87	3.04	-68.18
	(0.31)	(0.53)	(0.44)	(0.97)
Yield spreads	7.52	-0.22	-28.13	-60.01
	(0.30)	(0.50)	(0.88)	(0.97)
Common factor based forecasters				
Initial	2.07	-9.70	-24.39	-115.71
	(0.45)	(0.64)	(0.84)	(1.00)
Recursive	-1.86	-14.47	-17.27	-127.99
	(0.55)	(0.71)	(0.74)	(0.99)
Time series models				
HAR	26.48	22.79	6.13	-9.99
	(0.08)	(0.19)	(0.28)	(0.72)
Mean-reverting realized GARCH	11.87	4.89	-31.63	-88.66
	(0.22)	(0.42)	(0.92)	(1.00)

This table displays R^2_{OoS} measures in percent relative to a RW for all forecasting methods and maturities. In parentheses asymptotic p-values for a one-sided Diebold-Mariano test using the Newey-West variance estimator with automatic lag selection of Andrews (1991). For common factor based forecasters, the label "Initial" indicates that the selection of PCA factors is based on the initial estimation period, and "Recursive" that it is updated every period. The initial estimation period ranges from January 2, 2000, through June 6, 2008, and the out-of-sample period from June 7, 2008, through December 31, 2020.

C.4. The arbitrage-free Nelson-Siegel model with stochastic volatility

The $AFNS_3$ stochastic (or state-dependent) volatility model of Christensen et al. (2010) is an $A_3(3)$ model. Under $\mathbb{Q}$, the dynamics of the state variables are restricted to

$$dX_t = \begin{pmatrix} \varepsilon & 0 & 0 \\ 0 & \lambda & -\lambda \\ 0 & 0 & \lambda \end{pmatrix} \left[\begin{pmatrix} \tilde{\theta}_1 \\ \tilde{\theta}_2 \\ \tilde{\theta}_3 \end{pmatrix} - X_t \right] dt + \begin{pmatrix} \sigma_{1,1} & 0 & 0 \\ 0 & \sigma_{2,2} & 0 \\ 0 & 0 & \sigma_{3,3} \end{pmatrix} \begin{pmatrix} \sqrt{X_{1,t}} & 0 & 0 \\ 0 & \sqrt{X_{2,t}} & 0 \\ 0 & 0 & \sqrt{X_{3,t}} \end{pmatrix} dW_t^{\mathbb{Q}}.$$

The Riccati equations (6) are then given by

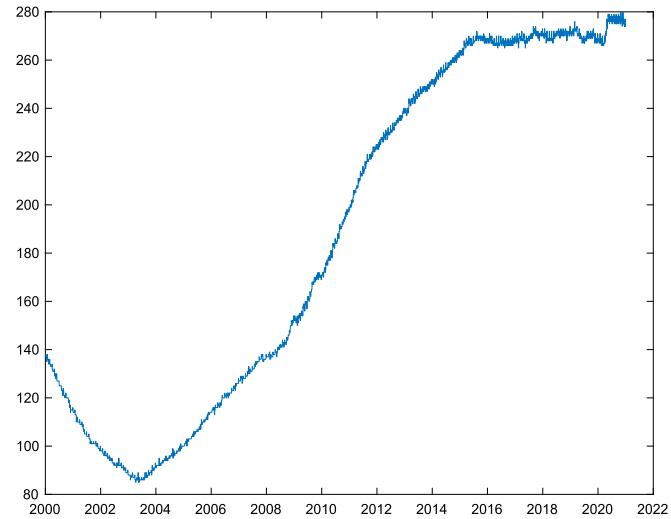


Fig. A.2. Number of coupon bonds over time.
This figure shows the evolution in the number N_t of coupon bonds with maturity between 3 months and 10 years used in the daily cross-sectional estimations Eq. (33). Callable issues are excluded. The sample spans the period from January 2, 2000, through December 31, 2020.

$$dB_1(\tau) = 1 - \varepsilon B_1(\tau) - \frac{1}{2}\sigma_{1,1}^2,$$
$$dB_2(\tau) = 1 - \lambda B_2(\tau) - \frac{1}{2}\sigma_{2,2}^2,$$
$$dB_3(\tau) = \lambda B_2(\tau) - \lambda B_3(\tau) - \frac{1}{2}\sigma_{3,3}^2,$$
$$dA(\tau) = B(\tau)'\tilde{\kappa}\tilde{\theta}.$$

We solve the ODEs numerically for each trial parameter vector in the iterative estimation procedure. For $\varepsilon \to 0$, the yield curve converges to Nelson and Siegel (1987) shape, except for the modification by $\frac{A(\tau)}{\tau}$, as in Eq. (C.5). Following Christensen et al. (2010), we set $\varepsilon = 10^{-6}$, restrict κ to be diagonal, and adopt the extended affine specification for the market price of risk. To ensure that the factors stay positive, we impose the Feller condition under both probability measures.

Appendix D. Estimation procedure

The term structure models are estimated using the Kalman filter, following Duffee (2002), Christensen et al. (2011), and others. Let y_t, $t = 1, \ldots, T$, be the N-vector of observed yields, with t counting the time increments between observations of $\Delta = 1/250$ years, $\tilde{A} = \left(\frac{A(\tau_1)}{\tau_1}, \ldots, \frac{A(\tau_N)}{\tau_N}\right)'$, and $\tilde{B}$ the $d \times N$ matrix with columns $\frac{B(\tau_i)}{\tau_i}$. By Eqs. (18) and (19), the state space system is thus given by the measurement and transition equations

$$y_t = \tilde{A} + \tilde{B}' X_t + \varepsilon_t, \tag{D.6}$$
$$X_t = C_\Delta + D_\Delta' X_{t-1} + \eta_t, \tag{D.7}$$

where

$$\begin{pmatrix} \varepsilon_t \\ \eta_t \end{pmatrix} \sim \mathcal{N}\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} H_\Delta & 0 \\ 0 & Q_{t,\Delta} \end{pmatrix} \right].$$

The conditional expectation and variance of the state process are given by

$$\mathbb{E}_{t-1}(X_t) = (I - \exp(-\kappa\Delta))\theta + \exp(-\kappa\Delta)X_{t-1} = C_\Delta + D_\Delta' X_{t-1}, \tag{D.8}$$

$$Var_{t-1}(X_t) = \int_{t-1}^{t} \exp(-\kappa(t-u))\sigma(\mathbb{E}_{t-1}(X_u))\sigma(\mathbb{E}_{t-1}(X_u))' \exp(-\kappa(t-u))' du = Q_{t,\Delta}, \tag{D.9}$$

where $\exp(\cdot)$ is the matrix exponential, and $\sigma(\cdot) = \Sigma\sqrt{S(\cdot)}$, with $S(\cdot)$ from Eq. (3). For example, $\sigma(X_u) = \mathrm{diag}(\sigma_{i,i}\sqrt{X_{i,u}})$ in the model in Appendix C.4. By Eq. (D.8), the transition parameters in Eqs. (19) and (D.7) are $C_\Delta = (I_d - \exp(-\kappa\Delta))\theta$ and $D_\Delta = \exp(-\kappa\Delta)$. Write $Y_t = (y_1, \dots, y_t)$ for the data through t and define $X_{t|t-1} = \mathbb{E}(X_t \mid Y_{t-1})$, $\Sigma_{t|t-1} = Var(X_t \mid Y_{t-1})$, $X_{t|t} = \mathbb{E}(X_t \mid Y_t)$, and $\Sigma_{t|t} = Var(X_t \mid Y_t)$. Suppressing Δ for notational ease, the prediction step is then

$$X_{t|t-1} = C + D' X_{t-1|t-1}\,, \tag{D.10}$$

$$\Sigma_{t|t-1} = D'\Sigma_{t-1|t-1} D + Q_t\,, \tag{D.11}$$

where Q_t is the conditional variance of the state variables from Eq. (D.9), while $D'\Sigma_{t-1|t-1}D$ reflects conditional mean variation, cf. footnote 3. The update step is

$$X_{t|t} = X_{t|t-1} + \Sigma_{t|t-1}\tilde{B}F_t^{-1}v_t\,, \tag{D.12}$$

$$\Sigma_{t|t} = \Sigma_{t|t-1} - \Sigma_{t|t-1}\tilde{B}F_t^{-1}\tilde{B}'\Sigma_{t|t-1}\,, \tag{D.13}$$

with the one step ahead prediction error and its variance given by

$$v_t = y_t - \tilde{A} - \tilde{B}' X_{t|t-1}\,, \tag{D.14}$$

$$F_t = Var(v_t) = \tilde{B}'\Sigma_{t|t-1}\tilde{B} + H\,. \tag{D.15}$$

The filter is initiated by setting $X_{0|0}$ and $\Sigma_{0|0}$ equal to the unconditional expected value and variance. From Eq. (2), $X_{0|0} = \theta$, and from Eq. (D.7), $\Sigma_{0|0} = D'\Sigma_{0|0}D + Q$, with Q from Eq. (D.9) at $\mathbb{E}_{t-1}(X_u) = \theta$. Thus, $vec(\Sigma_{0|0}) = \left(I_{d^2} - D' \otimes D'\right)^{-1} vec(Q)$.

Given the Gaussianity assumptions on measurement and transition equations, the parameters ψ are estimated by maximizing the prediction error decomposition of the conditional log likelihood function,

$$L(\psi) = \sum_{t=1}^{T}\left(-\frac{1}{2}N\log(2\pi) - \frac{1}{2}\log\left(\det F_t\right) - \frac{1}{2}v_t' F_t^{-1} v_t\right). \tag{D.16}$$

The states X_t are Gaussian for the Vasicek and AFNS$_0$ models. For the square root processes (the CIR and AFNS$_3$ models), estimation based on Eq. (D.16) amounts to QML. Here, the Gaussian approximation implies that states are not restricted to be positive. In this case, following Chen and Scott (2003), we truncate states at 0. Given the large number of parameters in some of the models, we use the global optimizer differential evolution with several starting values. For every 250 observations, we reestimate parameters using differential evolution. In intermediate periods we use local optimizers.

D.1. Predictive regression correction

From Eq. (D.6), $var(y_t \mid X_{t-1}) = \tilde{B}' Var_{t-1}(X_t)\tilde{B} + H$, suppressing dependence on the time increment Δ. Using this and Eqs. (D.9), (D.11) and (D.15),

$$\begin{aligned} Var(y_t \mid Y_{t-1}) &= Var(v_t) \\ &= \tilde{B}'(D'\Sigma_{t-1|t-1}D + Var_{t-1}(X_t))\tilde{B} + H \\ &= \tilde{B}' D'\Sigma_{t-1|t-1}D\tilde{B} + Var(y_t \mid X_{t-1})\,. \end{aligned} \tag{D.17}$$

Thus, the conditional variance of yields is given by that corresponding to perfect observation of state variables through $t-1$, i.e., the second term in Eq. (D.17), with an adjustment for conditional mean variation due to imperfect state observation given by the first term in Eq. (D.17) (see footnote 3). When forecasting using the predictive regression in Eq. (1), with Z_t given by Eq. (7), then Z_t corresponds to the second term in Eq. (D.17). Forecasting therefore involves three modifications. First, the filtered state $X_{t|t}$ is used in place of X_t in Eq. (7). Second, using Eq. (1), $\rho_0^{\tau,h}$ and $\rho_1^{\tau,h}$ provide empirical predictive regression corrections for the bias stemming from conditional mean variation, i.e., the first term in Eq. (D.17). Finally, the estimated coefficients in Eq. (1) reflect the history of realized volatilities, which are part of investor's information set, thereby conditioning on a larger information set than that based on daily yields in the Kalman filter.

D.2. Conditional moments of state process

By the results in Fackler (2000), also used in Jacobs and Karoui (2009), the first two conditional moments can be written as

$$\mathbb{E}_t(X_{t+h}) = C_h + D_h' X_t\,, \tag{D.18}$$

$$vec(Var_t(X_{t+h})) = \zeta_{0,h} + \zeta_{1,h}' X_t\,, \tag{D.19}$$

with $C_h = (I_d - \exp(-\kappa h))\theta$ and $D_h = \exp(-\kappa h)$, see Eqs. (D.8)-(D.9). The $d^2 \times 1$ vector of intercepts $\zeta_{0,h}$ in Eq. (D.19) is determined by

$$\begin{pmatrix} C_h \\ \zeta_{0,h} \end{pmatrix} = (I_{d+d^2} - \exp(\Pi h))\Pi^{-1}\Theta\,, \tag{D.20}$$

and the $d \times d^2$ matrix of slope terms $\zeta_{1,h}$ by

$$\begin{pmatrix} D_h \\ \zeta_{1,h}' \end{pmatrix} = \exp(-\Pi h)\begin{bmatrix} I_d \\ 0 \end{bmatrix}. \tag{D.21}$$

In Eq. (D.20), the $(d+d^2)$-vector Θ is

$$\Theta = \begin{bmatrix} \kappa\theta \\ (\Sigma \otimes \Sigma)\mathcal{D}\alpha \end{bmatrix},$$

with $\alpha = 0$ in Eq. (3) for the models we consider, and $\mathcal{D}$ given by

$$\mathcal{D}_{i,j} = \begin{cases} 1 & \text{if } i = (j-1)d + j\,, \\ 0 & \text{otherwise}\,. \end{cases}$$

Further,

$$\Pi = \begin{bmatrix} \kappa & 0 \\ -(\Sigma \otimes \Sigma)\mathcal{D}\mathcal{B} & \kappa \otimes I_d + I_d \otimes \kappa \end{bmatrix},$$

a $(d+d^2)\times(d+d^2)$ matrix, with the ith row of $\mathcal{B}$ given by β_i' from Eq. (3), and $\operatorname{rank}\mathcal{B} = m \le d$.

In the special case $m = 0$, i.e., an $A_0(d)$ model, we have $\mathcal{B} = 0$ and, hence,

$$\Pi = \begin{bmatrix} \kappa & 0 \\ 0 & \kappa \otimes I_d + I_d \otimes \kappa \end{bmatrix}.$$

From Eq. (D.21),

$$\begin{pmatrix} D_h \\ \zeta_{1,h}' \end{pmatrix} = \begin{pmatrix} \exp(-\kappa h) \\ 0 \end{pmatrix} \tag{D.22}$$

in this case. In particular,

$$\zeta_{1,h} = 0\,, \text{ for } m = 0\,. \tag{D.23}$$

D.3. Conditional moments of yields

By Eqs. (5) and (D.18), the conditional means of the yields are

$$\mathbb{E}_t(y_{t+h}) = \tilde{A} + \tilde{B}'\left(C_h + D_h' X_t\right). \tag{D.24}$$

Using Eqs. (7) and (D.19), the conditional yield variances are

$$\begin{aligned} Var_t(y_{t+h}^\tau) &= \frac{1}{\tau^2}B(\tau)' Var_t(X_{t+h})B(\tau) \\ &= \frac{1}{\tau^2}(B(\tau)\otimes B(\tau))' vec(Var_t(X_{t+h})) \\ &= b_0^{\tau,h} + b_1^{\tau,h\prime} X_t, \end{aligned} \tag{D.25}$$

$$b_0^{\tau,h} = \frac{1}{\tau^2}(B(\tau)\otimes B(\tau))'\zeta_{0,h}\,, \tag{D.26}$$

$$b_1^{\tau,h\prime} = \frac{1}{\tau^2}(B(\tau)\otimes B(\tau))'\zeta_{1,h}'\,, \tag{D.27}$$

where $b_0^{\tau,h}$ is a scalar, and $b_1^{\tau,h}$ a $d \times 1$ vector.

In the special case $m = 0$, i.e., an $A_0(d)$ (Gaussian) model, such as Vasicek or AFNS$_0$, we have from Eq. (D.23) that $\zeta_{1,h} = 0$. Thus, by Eq. (D.27), it follows that $b_1^{\tau,h\prime} = 0$ and, by Eq. (D.25),

$$Var_t(y_{t+h}^{\tau}) = b_0^{\tau,h}, \text{ for } m = 0. \tag{D.28}$$

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